MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
PRIMARY CRUSHING
MODIFICATION OF CRUSHER SUPPORT STRUCTURE

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PREFACE

VIBRATION PROBLEM - PRIMARY CRUSHER & SUPPORTING STRUCTURE

A. The Problem:

- At 100% speed of crusher with full load, vibration was being transmitted to the supporting concrete structure and was felt.
- Vibration measurement was taken and maximum observed value in vertical direction was 3.3 mm/s (RMS), in horizontal direction was 6.9 mm/s (RMS) along the conveyor direction, and 3.1 mm/s (RMS) across the conveyor direction. DIN 4150 specifies an allowable limit of 20 mm/s for industrial & commercial building and 3mm/s for special/ sensitive structure. However, as per project specification allowable limit is only 3 mm/s. Manufacturer has not specified any allowable limit of vibration.

B. Root Cause Analysis :

Static analysis of the Primary Crusher building was carried out in Staad Software during execution stage along with Modal analysis to check the frequency separation from the exciting frequency of the Crusher. The root cause can be as follows:

- Misinterpretation of the forced vibration nature of the equipment.
- Incompliance with project requirements with regard to being away from the range of excitation frequencies and vibration amplitude limit; and
- Misinterpretation of the geotechnical advice with regard to foundation design parameters for dynamic foundations.

C. Immediate Remedial Action Proposed:

- Identifying the possible problems in the machine, and rectifying the same is a time taking activity.
- Under the circumstances, we propose to stiffen the structure, to the maximum extent, by providing concrete walls between columns, wherever possible.
- This stiffening should significantly increase the Natural frequency of the structure and should reduce the amplitude of vibrations.
- Based on this consideration, a detailed analysis (Eigen value, Frequency Response Function) has been carried out in full scale 3D, using ANSYS. The calculated Natural Frequency and Amplitude of vibration are as per requirement and should ensure safe operation of the machine.

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D. Conclusion:

Strengthening at few locations (Appendix 1) using additional walls will eliminate any possible vibrations in the Crusher floor. After strengthening:

- the computed RMS vibration velocity is 2.667 mm/s, which is within the project specified limit.
- the relevant natural frequency has been high-tuned to 2 times of working frequency
- the modified structure is adequate for the loads arising from the equipment
- structural strength along with base pressure has been checked and found to be with in project specification safe limit [4].

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1. Introduction

This report is a verification of the Primary Crusher system, subject to a range of sinusoidal loading conditions using finite element analysis [1]. Two set of analyses are performed with empty and loaded condition of the crusher.

The main purpose of this analysis is to investigate:

- Undamped modal behavior to check with the excitation frequencies of the Primary Crusher.
- Steady state dynamic behavior of the system with harmonic analysis to ensure that potential excitation forces will not cause vibration that might jeopardize safe operation of the equipment.

Two sets of modal & harmonic analyses are performed considering empty and loaded conditions of the crusher.

2. Design Basis

This report is only for dynamic analysis, and separate report was created for civil design [4].

2.1. Geometry

Primary Crusher system consists of the Soils, RCC footing, columns, beams, ties, slab, and different pedestals along with steel superstructure shown in Figure 1. Dimension of the soil model is considered as 45m X 45m X 15m depth.

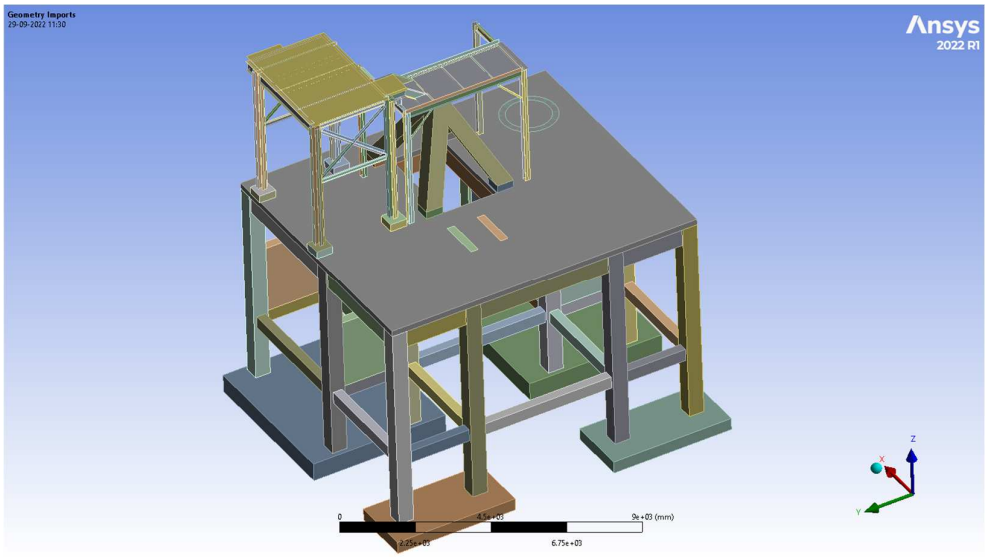


Figure 1a. ISO View Primary Crusher Structure

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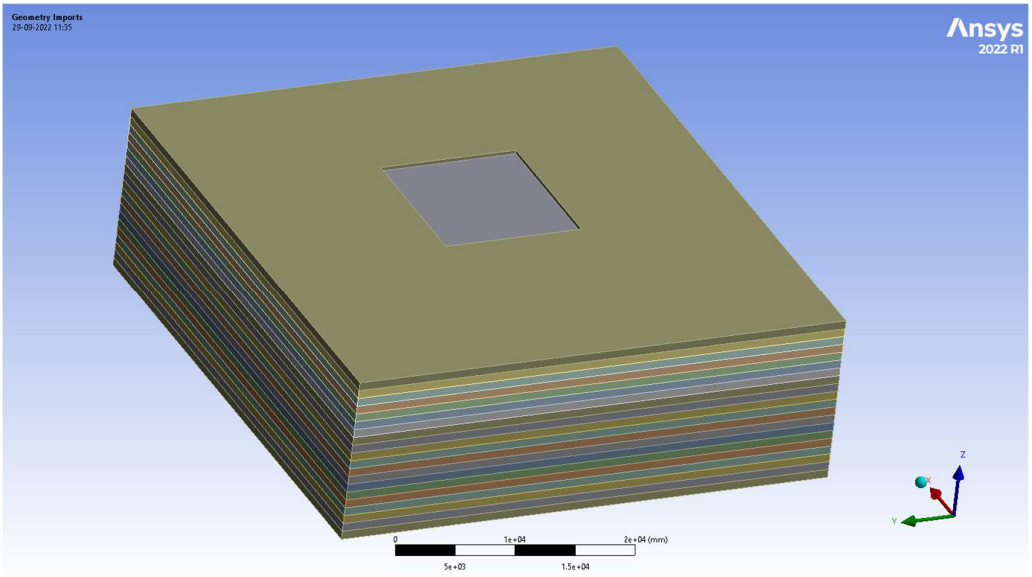


Figure 1b. Soil model of Primary Crusher System

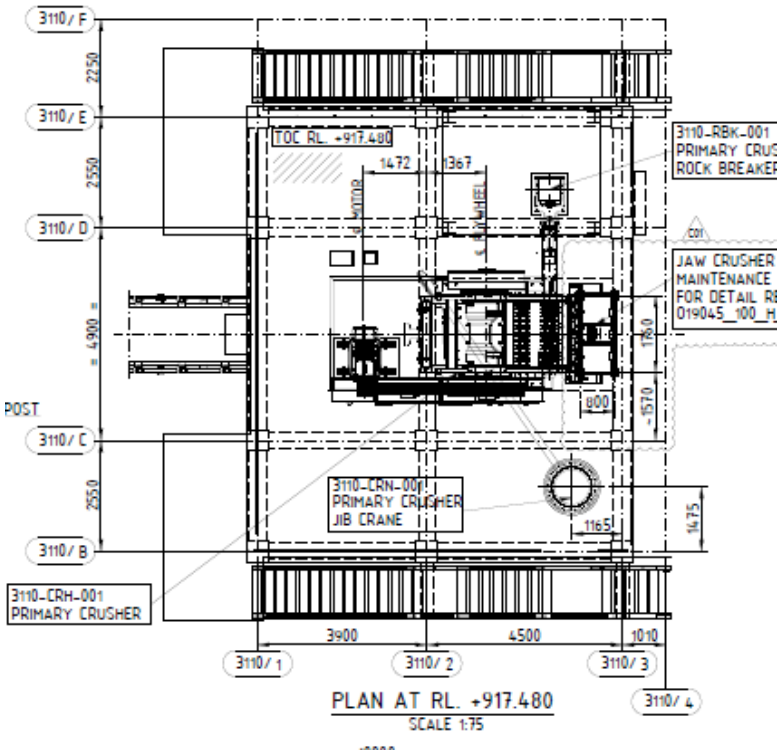


Figure 1c. Plan dimensions of Primary Crusher System

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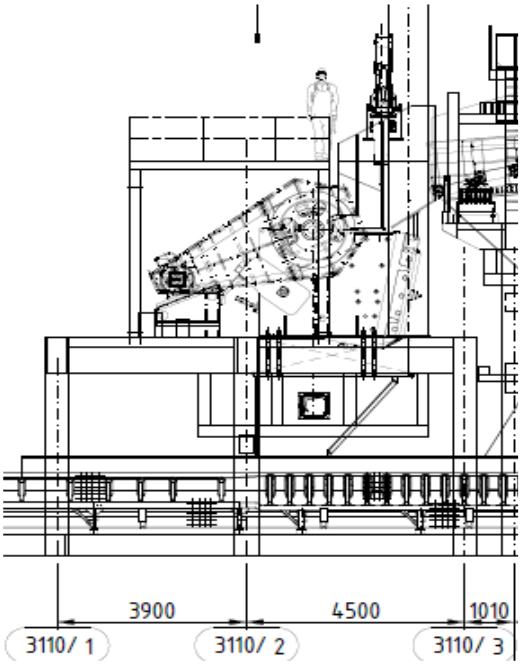


Figure 1d. Elevation Dimensions of Primary Crusher System

2.2. Material Properties

The material properties of concrete shown in Table 1a after adjusting vibrational data from measurement.

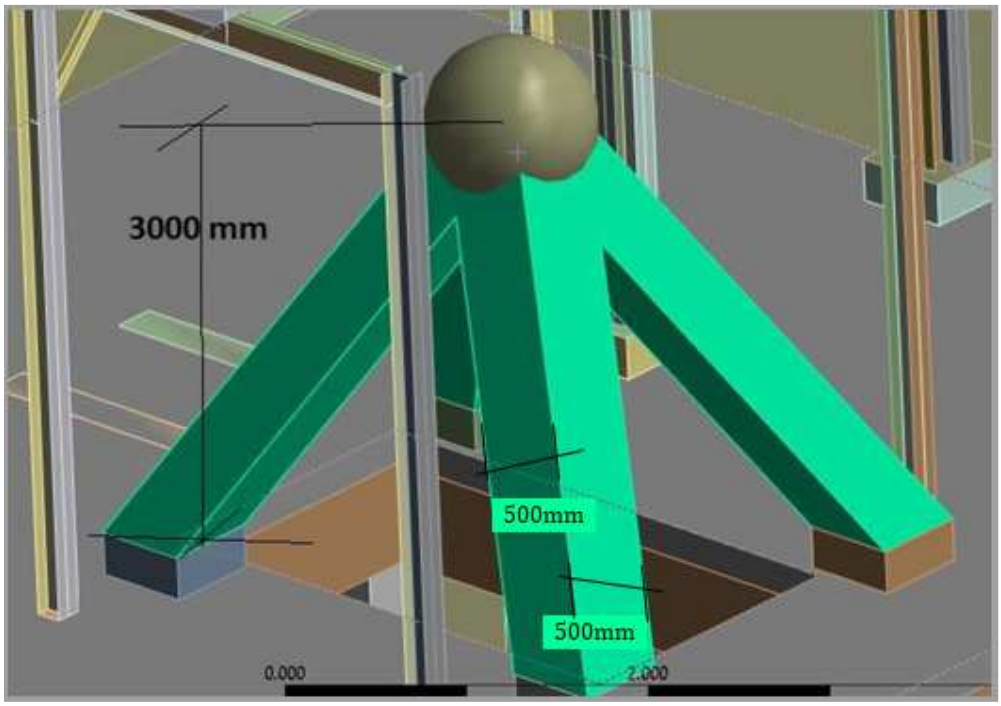
Property	Value	Unit
Material Field Variables	Table	
Density	2400	kg m^-3
Isotropic Secant Coefficient of Thermal Expansion		
Isotropic Elasticity		
Derive from	Young's Mod...	
Young's Modulus	3.1528E+10	Pa
Poisson's Ratio	0.17	
Bulk Modulus	1.5923E+10	Pa
Shear Modulus	1.3474E+10	Pa

Table 1a. Material properties of concrete

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2.3. Rigid Link

Pedestal top to Center of machine is connected with massless rigid links to mimic the COG of eccentric shaft as shown in Appendix 2.



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2.4. Soil Properties

Borehole 19 is near Crusher System. Shear modulus & Poisson ratios are taken from the soil report to model the soil in different layers. The material properties of soil in Figure 2.

TEST REPORT - DOWNHOLE SEISMIC TEST RESULTS *(ASTM D7400 - 08)*

FORM # RGF-KHO-GEO-DH-1-00

Project: *MANSOURAH MASSARAH GOLD PROJECT*
Client: *L & T CONSTRUCTION*
Project No.: *19-K-3054* **Test No.** *DH-03*
Borehole No.: *BH-19* **Geotechnical File No.:** -
Source Type: *Sledge Hammer* **Source to BH Distance (m):** 3 m
Receiver Type: *Triaxial Geophones* **Coordinates (m):** *N: 2506510.84, E: 264424.58*

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density ρ (g/cm³)	Poisson Ratio υ	Shear Modulus G (M Pa)	Young Modulus E (M Pa)	Bulk Modulus K (M Pa)
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)					
1	0.75	3.09	5.10	9.09	606	340	1.7	0.27	197	500	363
2	1.50	3.35	5.38	9.56	935	557	1.7	0.22	527	1292	783
3	2.25	3.75	5.78	10.21	990	609	1.8	0.20	668	1596	873
4	3.00	4.24	6.28	11.01	985	616	1.8	0.18	683	1610	837
5	3.75	4.80	6.87	11.97	949	583	1.8	0.20	612	1464	804
6	4.50	5.41	7.48	12.93	993	631	1.8	0.16	717	1666	820
7	5.25	6.05	8.16	14.04	939	575	1.8	0.20	595	1428	793
8	6.00	6.71	8.81	15.09	1018	630	1.8	0.19	714	1699	912
9	6.75	7.39	9.44	16.15	1077	640	1.8	0.23	737	1809	1104
10	7.50	8.08	9.94	16.97	1382	843	1.9	0.20	1350	3250	1830
11	8.25	8.78	10.26	17.50	2190	1322	2.1	0.21	3671	8908	5176
12	9.00	9.49	10.59	18.04	2146	1312	2.2	0.20	3785	9099	5089
13	9.75	10.20	10.90	18.55	2304	1401	2.2	0.21	4315	10417	5926
14	10.50	10.92	11.22	19.08	2247	1357	2.2	0.21	4050	9825	5709
15	11.25	11.64	11.58	19.66	2008	1246	2.2	0.19	3418	8113	4315
16	12.00	12.37	11.90	20.18	2269	1397	2.2	0.20	4291	10256	5609
17	12.75	13.10	12.23	20.74	2209	1302	2.2	0.23	3727	9198	5763
18	13.50	13.83	12.56	21.26	2216	1406	2.2	0.16	4349	10115	5000
19	14.25	14.56	12.96	21.89	1833	1164	2.2	0.16	2979	6924	3417
20	15.00	15.30	13.42	22.63	1597	993	2.2	0.19	2169	5140	2721
21	15.75	16.03	13.87	23.39	1636	969	2.2	0.23	2064	5077	3135
22	16.50	16.77	14.26	24.02	1891	1170	2.2	0.19	3014	7168	3846

Figure 2a. Soil Properties for different layers

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Properties of Outline Row 11: Rock1

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1700	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.0038E+08	Pa
9	Poisson's Ratio	0.27	
10	Bulk Modulus	3.6259E+08	Pa
11	Shear Modulus	1.97E+08	Pa

Properties of Outline Row 22: Rock2

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1700	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.2859E+09	Pa
9	Poisson's Ratio	0.22	
10	Bulk Modulus	7.654E+08	Pa
11	Shear Modulus	5.27E+08	Pa

Properties of Outline Row 26: Rock3

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6032E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	8.9067E+08	Pa
11	Shear Modulus	6.68E+08	Pa

Properties of Outline Row 27: Rock4

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6119E+09	Pa
9	Poisson's Ratio	0.18	
10	Bulk Modulus	8.3952E+08	Pa
11	Shear Modulus	6.83E+08	Pa

Figure 2b. Properties of soil layers 1 to 4

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Properties of Outline Row 28: Rock5

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.4688E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	8.16E+08	Pa
11	Shear Modulus	6.12E+08	Pa

Properties of Outline Row 29: Rock6

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6634E+09	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	8.1541E+08	Pa
11	Shear Modulus	7.17E+08	Pa

Properties of Outline Row 30: Rock7

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.428E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	7.9333E+08	Pa
11	Shear Modulus	5.95E+08	Pa

Properties of Outline Row 31: Rock8

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6993E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	9.1361E+08	Pa
11	Shear Modulus	7.14E+08	Pa

Figure 2c. Properties of soil layers 5 to 8

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Properties of Outline Row 32: Rock9

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.813E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	1.1191E+09	Pa
11	Shear Modulus	7.37E+08	Pa

Properties of Outline Row 12: Rock10

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1900	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	3.24E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	1.8E+09	Pa
11	Shear Modulus	1.35E+09	Pa

Properties of Outline Row 13: Rock11

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2100	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	8.8838E+09	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	5.1056E+09	Pa
11	Shear Modulus	3.671E+09	Pa

Properties of Outline Row 14: Rock12

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.084E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	5.0467E+09	Pa
11	Shear Modulus	3.785E+09	Pa

Figure 2d. Properties of soil layers 9 to 12

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Properties of Outline Row 15: Rock13

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.0442E+10	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	6.0013E+09	Pa
11	Shear Modulus	4.315E+09	Pa

Properties of Outline Row 16: Rock14

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.801E+09	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	5.6328E+09	Pa
11	Shear Modulus	4.05E+09	Pa

Properties of Outline Row 17: Rock15

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	8.1348E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	4.3736E+09	Pa
11	Shear Modulus	3.418E+09	Pa

Properties of Outline Row 18: Rock16

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.0298E+10	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	5.7213E+09	Pa
11	Shear Modulus	4.291E+09	Pa

Figure 2e. Properties of soil layers 13 to 16

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Properties of Outline Row 19: Rock17

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.1684E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	5.6595E+09	Pa
11	Shear Modulus	3.727E+09	Pa

Properties of Outline Row 20: Rock18

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.009E+10	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	4.9459E+09	Pa
11	Shear Modulus	4.349E+09	Pa

Properties of Outline Row 21: Rock19

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	6.9113E+09	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	3.3879E+09	Pa
11	Shear Modulus	2.979E+09	Pa

Properties of Outline Row 23: Rock20

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.1622E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	2.7754E+09	Pa
11	Shear Modulus	2.169E+09	Pa

Figure 2f. Properties of soil layers 17 to 20

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Properties of Outline Row 24: Rock21

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.0774E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	3.1342E+09	Pa
11	Shear Modulus	2.064E+09	Pa

Properties of Outline Row 25: Rock22

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	7.1733E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	3.8566E+09	Pa
11	Shear Modulus	3.014E+09	Pa

Figure 2g. Properties of soil layers 21 to 22

2.5. Lump Mass Modelling

Mass model of the crusher, listed in Table 2a, 2b ,2c 2d & 2e is considered as equivalent amount mass of material for simplification of the model in finite element analysis. Modelling is done taking information from load data given by equipment manufacturer as follows:

Loaded Conditions:

- Mass of Fv [distributed equally at 4 positions]: 200kN = around 20,000 kg
- Mass of PC/2 : 45kN = around 4,500 kg

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	VERTICAL FORCE	HORIZONTAL LONGITUDINAL FORCE	HORIZONTAL TRANSVERSAL FORCE
	F_V	F_{II}	$F_{\perp}$
STATIC LOAD 1	200 kN		
DYNAMIC LOAD 2	± 31 kN	± 13 kN	± 3 kN
NOTE: THE DYNAMIC LOAD FREQUENCY IS 3.3 HZ FOR THE VERTICAL AND LONGITUDINAL LOAD USING THE 90 DEGREE METHOD SETTINGS. THE FORCES ARE DERIVED BASED ON A RIGID FOUNDATION ASSUMPTION.			

Table 2a. Load data from [MD - 216 -3110 - EG- ME -GAD -V500-1007 Rev C01] [2]

POINT	ELEVATION IN "m"	VERTICAL LOAD (IN "t")
PC/2	(+)916.985	(+)4.5

Table 2b. Load data from [MD - 216 -3110 - EG- ME -ELV - 1008 Rev C01]

Empty Conditions:

- Mass of Fv [distributed equally at 4 positions]: = around 13025 kg each
- Mass of PC/2 : 45kN = around 4,500 kg

TOTAL WEIGHT	: 52,100 kg
LENGTH	: 4,125 mm
WIDTH	: 3,000 mm
HEIGHT	: 3,330 mm

Table 2c. Load data from [MD - 216 -3110 - EG- ME -GAD -V500-1007 Rev C01] [2]

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Load from JIB Crane

PILLAR JIB CRANE
3110-CRN-001 PRIMARY CRUSHER JIB CRANE

QUANTITY	:	One NO.
CRANE CAPACITY	:	6300 Kg
BOOM RADIUS	:	6000 mm
OVERALL HEIGHT	:	10190 mm
INSTALLATION	:	OUTDOOR OPERATION–DUSTY ENVIRONMENT
CRANE RATING	:	HC2 S3 HD2
SUPPLY VOLTAGE	:	400V 3PH+E, 60Hz VOLTAGE (±10% & FREQUENCY ±5% VARIATION) & 4TH WIRE FOR EARTHING–4 WIRE SUPPLY
CONTROL VOLTAGE	:	48 V
DEGREE OF PROTECTION	:	IP 65
WEIGHT OF HOIST + TROLLEY	:	824 Kg
TOTAL WEIGHT OF THE JIB	:	4645 Kg

Table 2d. Load data from [MD - 216 -3110 - EG- ME -GAD -V500-1007 Rev C01] [3]

Load from Rock Breaker Supporting Structure

The Rock Breaker supporting structure has been modeled along with concrete structure and mass of Rock Breaker has been applied as per table 2e.

Mass from Rock Breaker :

TABLE-1: LOAD TABLE FOR ROCK BREAKER
3110-RBK-01

POINT	ELEVATION IN "m"	VERTICAL LOAD (V) IN "t"	HORIZONTAL TOWARDS JAW CRUSHER (H1) IN "t"	MOMENT ABOUT CENTER (M) IN "t-m"	SWING TORQUE (MK) IN "t-m"
RB/1	(+)921.985	(+)3.5	7.5	20.0	3.0
LOAD COMBINATION FOR ROCK BREAKER : 1. V+H1+M 2. V+H1+MK 3. V+MK					

Table 2e. Load data from [MD-216-3110-EG-ME-LPX-1001-C01]

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The material properties of lumped structures and other components are considered to get such masses and listed in Figure 3a, 3b, 3c ,3d & 3e.

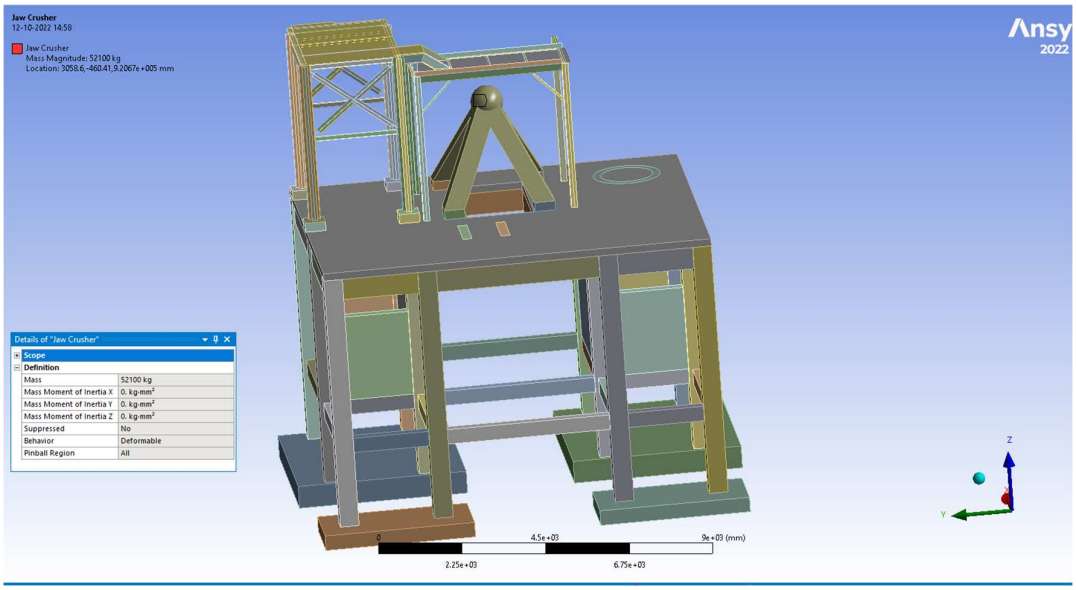


Figure 3a. Empty Lumped mass of the crushers at support

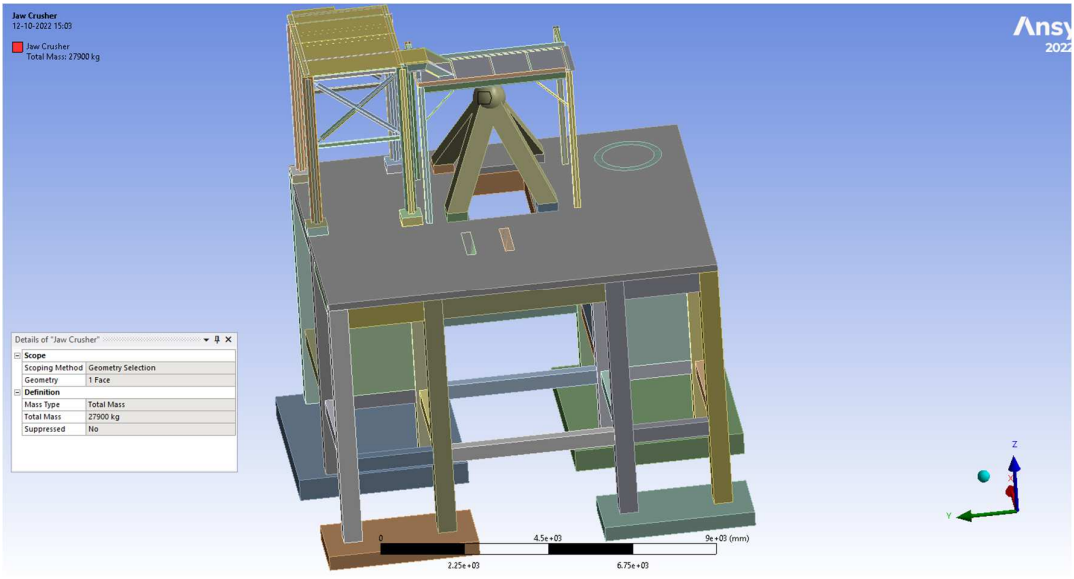
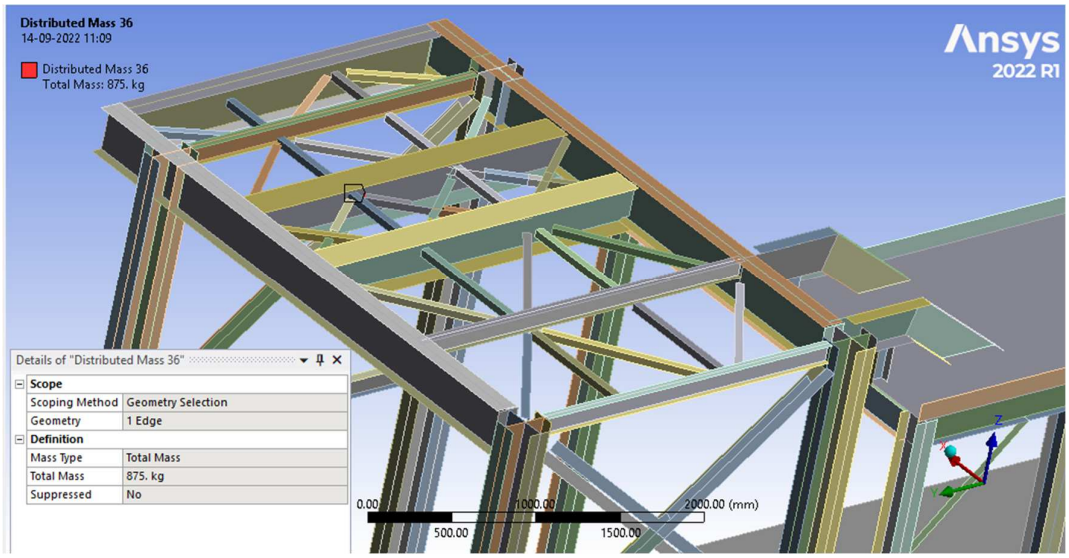


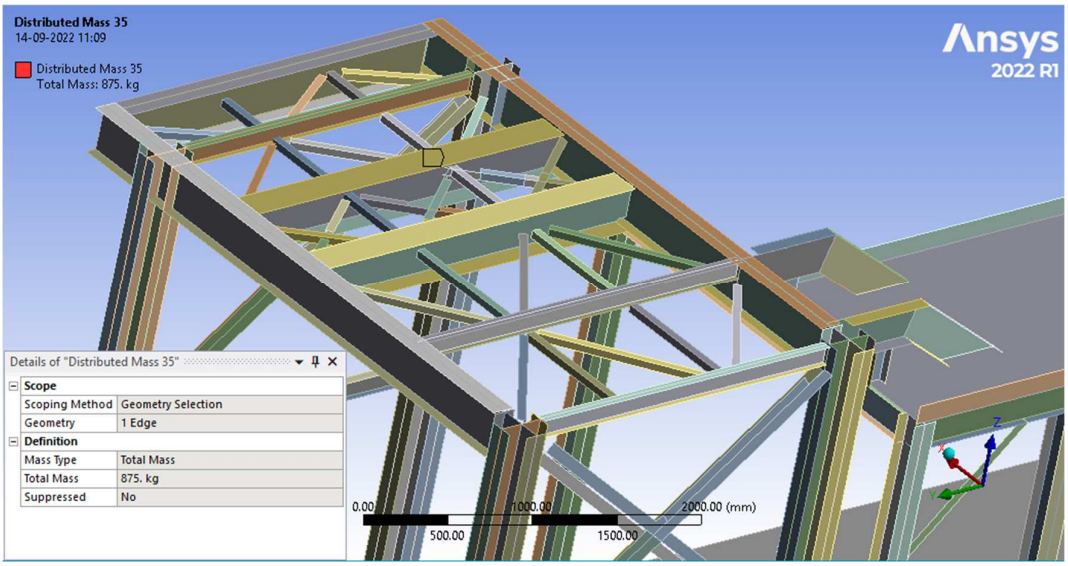
Figure 3b. Material Lumped mass of the crushers at support

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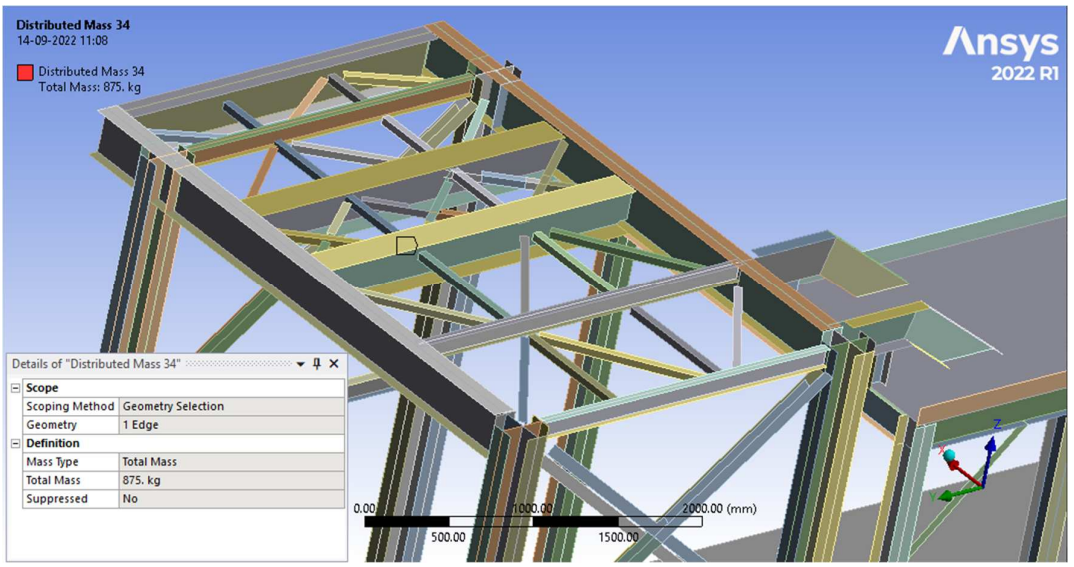




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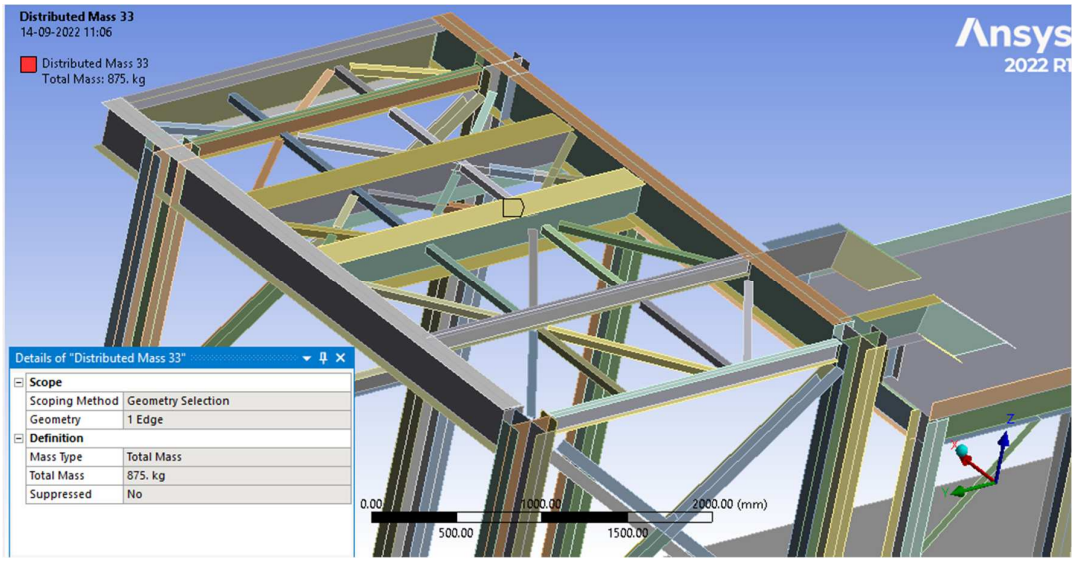


Figure 3c. Lumped mass of the Rock Breaker at support

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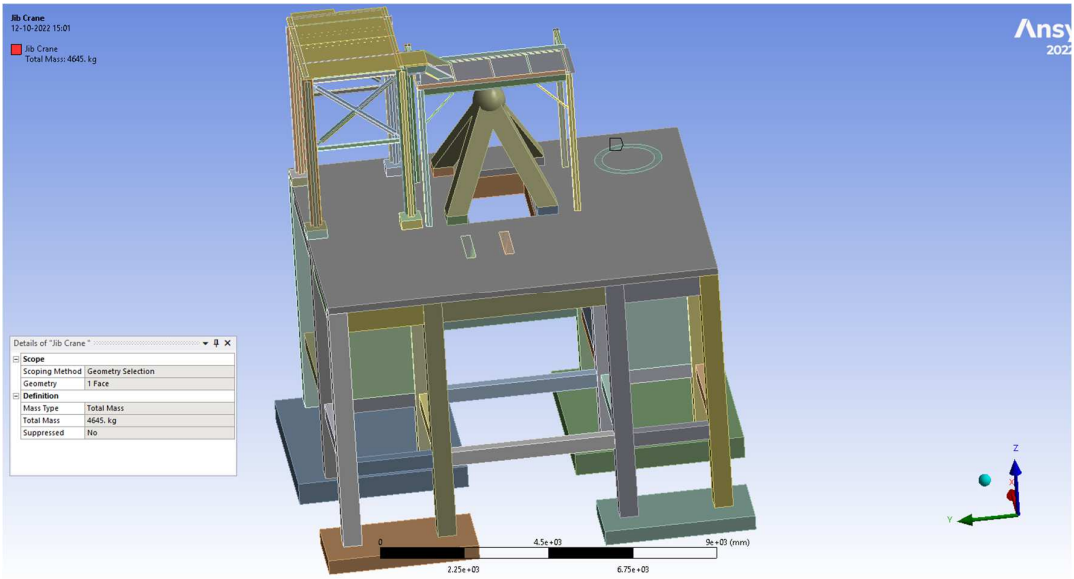
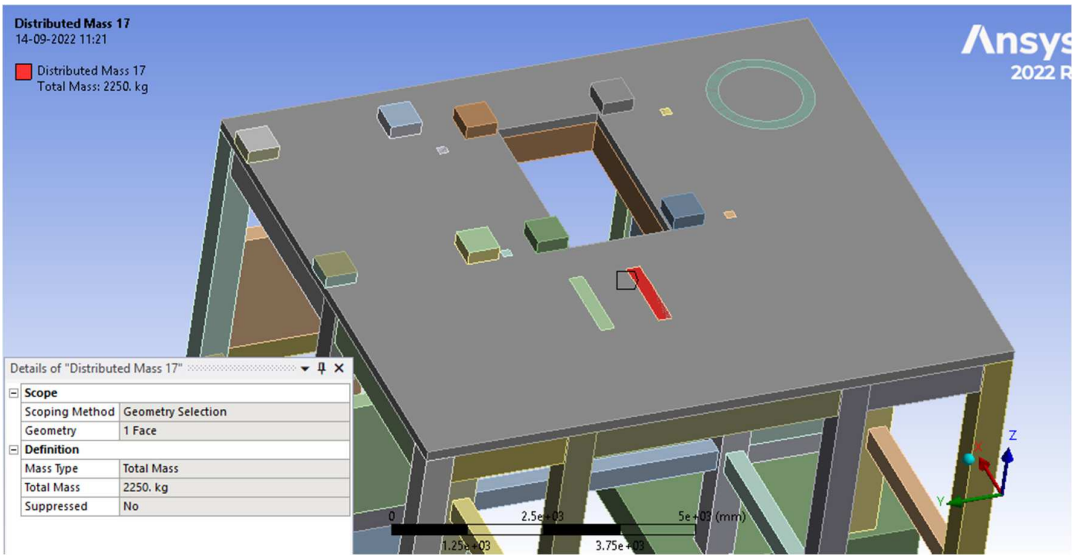


Figure 3d. Lumped mass of the Jib Crane at support



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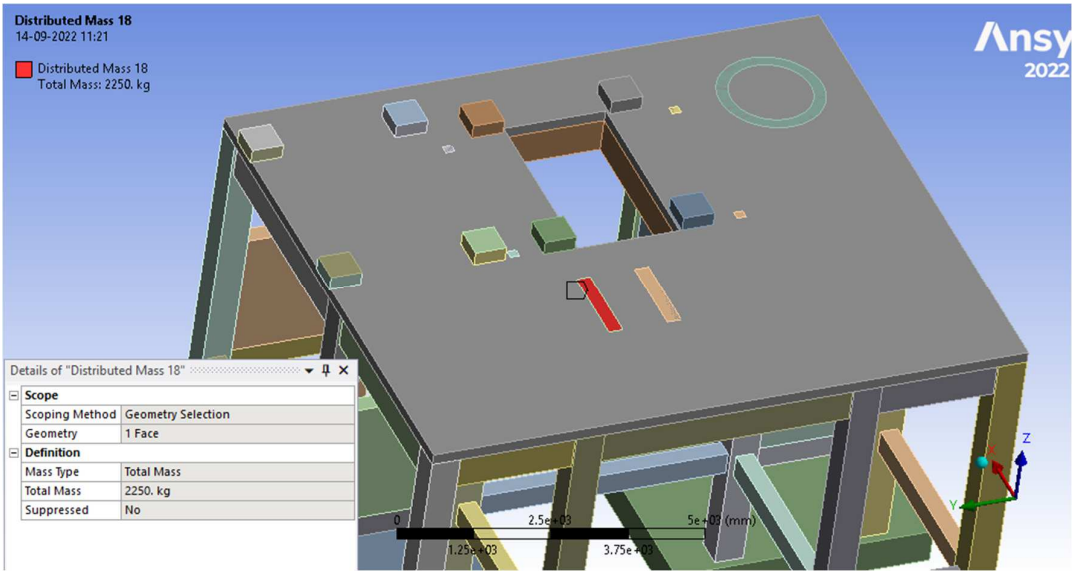


Figure 3e. Lumped mass of the Motor at support

2.6. Individual Component of the Model

Crusher is modelled with different components (shown in Figure-3f & 3g).

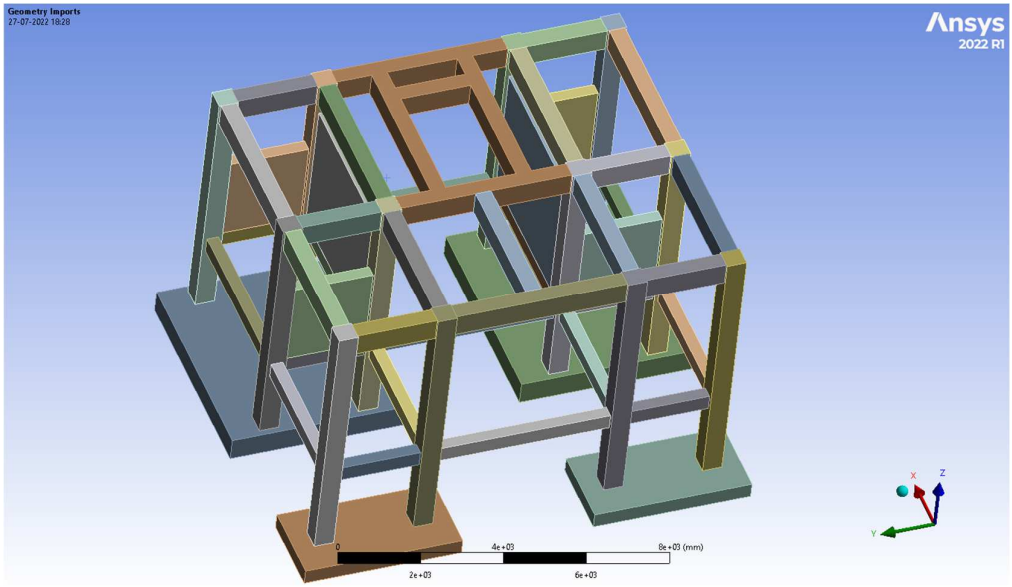


Figure 3f. Footings, beams, columns, & ties of the crusher system.

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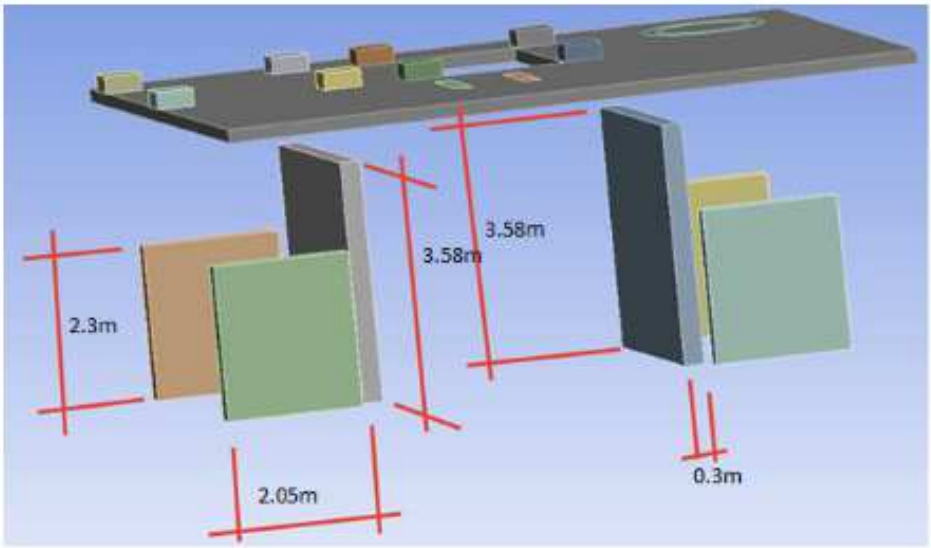


Figure 3g. slabs, pedestals, & proposed walls of the crusher system along with dimension.

2.7. Complete Model

Complete model of the system is shown in Figure 4.

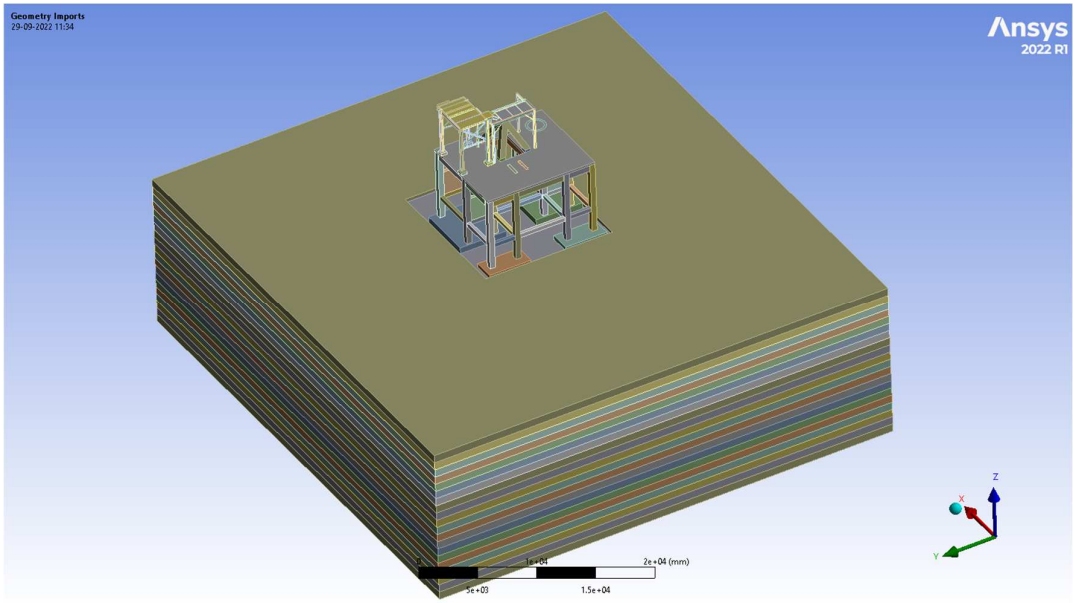


Figure 4a. Complete Model of the Primary Crusher system along with soil layers

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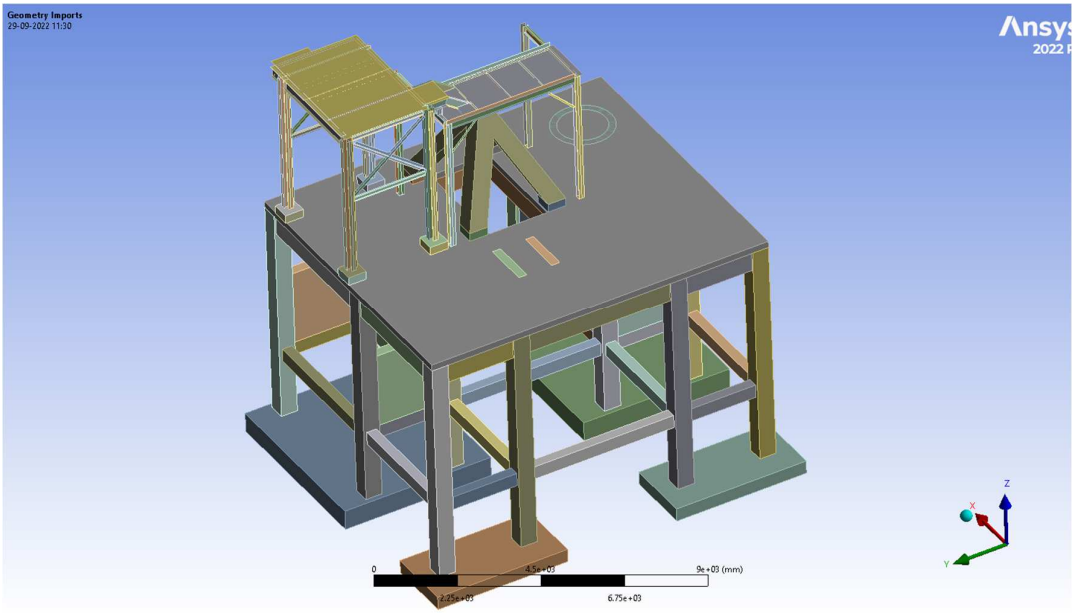


Figure 4b. Complete Model of the Primary Crusher system without soils

3. Finite Element Modelling

3.1. Mesh

Figures 5a & 5b shows the finite element meshes of the Concrete structure & Soil layers. The number of elements and nodes are listed in Table 3.

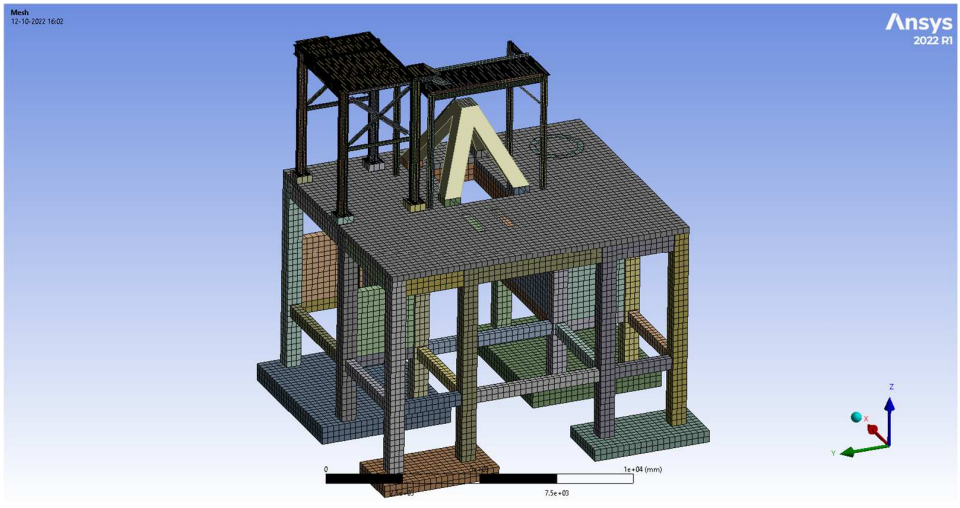


Figure 5a. Finite Element Mesh of Steel & Concrete structure

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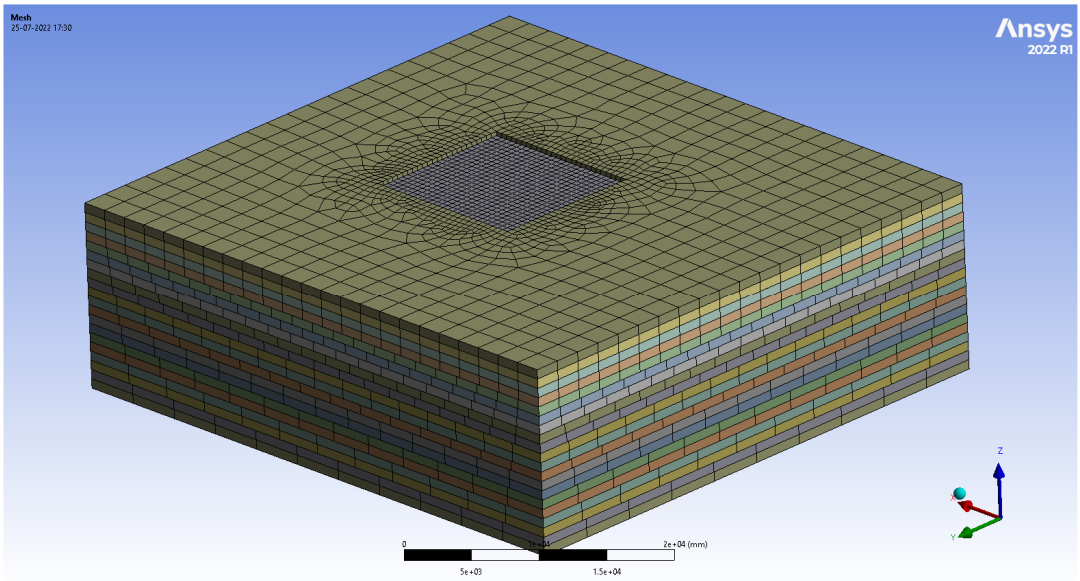


Figure 5b. Finite Element Mesh of soil layers

Statistics	
Nodes	138974
Elements	55070

Table 3. Finite Element Mesh Statistics

3.2. Boundary Conditions

The model is attached with respective boundary condition at the bottom & sides of the soil, shown in the Figure 6.

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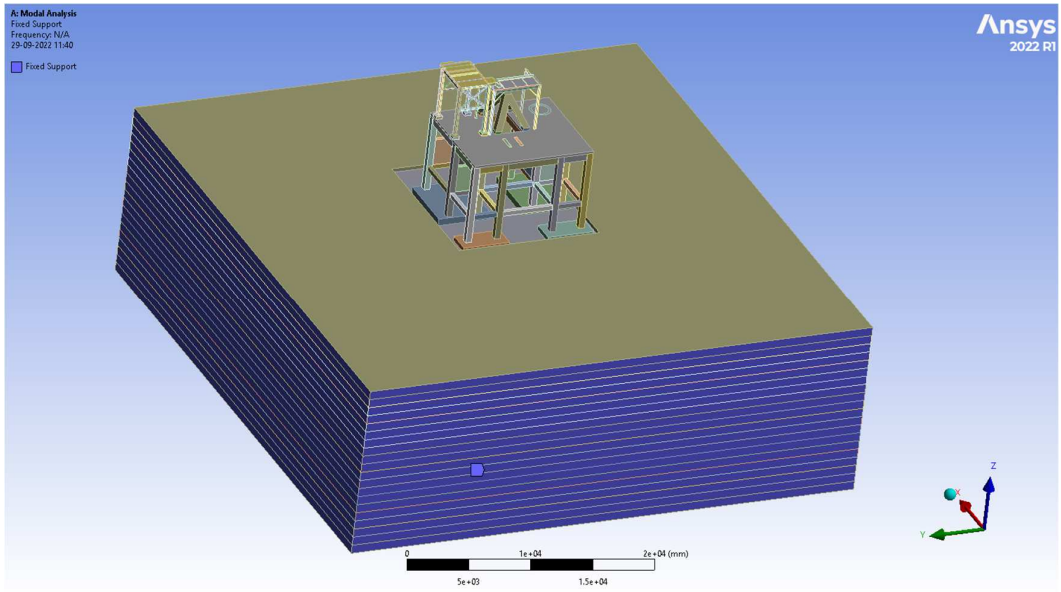


Figure 6. Boundary condition in the soil boundary

3.3. Dynamic Load, normal operation (f=3.3 Hz)

Components of dynamic forces are taken from drawing provided by equipment manufacturer. Continuous component (Static load) has no effect in dynamic analysis, so ignored in current analysis. Dynamic component is applied in the respective pedestals of the system with different phases and frequency (3.3 Hz). One set of analyses have been done due to excitation forces as shown as condition 1. Corresponding analysis results are shown for the conditions.

3.4. Philosophy of the Dynamic Loadings

Following analysis has been performed:
Condition 1: All horizontal forces are in the same phase. Vertical forces are 90 degree phase lag.
Typical forcing mechanisms are shown in Appendix 2.

3.5. Directions of the Loadings

Following harmonic loads are given at Centre of Gravity of Crusher.

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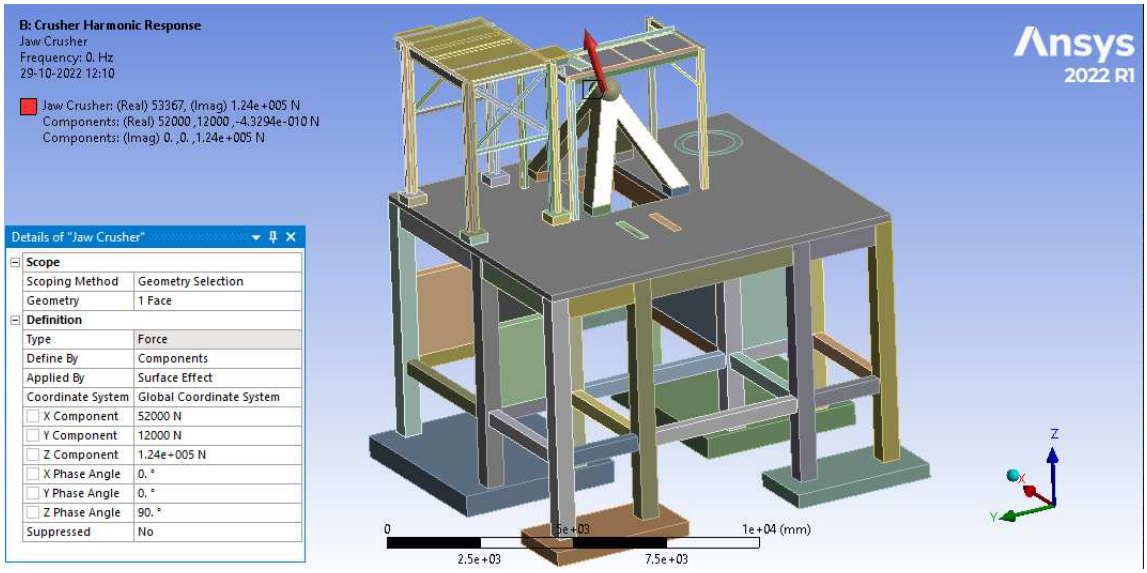


Figure 4a. Excitation Force : Condition 1

4. Analysis

Main purpose of the study is to perform modal & harmonic analysis.

4.1. Modal Analysis

A modal analysis is performed to determine the natural frequencies and mode shapes of the system. The purpose is to investigate if any natural modes would occur at potential excitation frequencies such as the crusher frequency.

4.2. Harmonic Analysis

Harmonic analysis is performed to determine the steady-state response of the structure subject to the dynamic crusher loads. The main purpose is to investigate if the natural modes identified in section 4.1 would result in vibration levels exceeding certain limit when subject to the dynamic load. Damping of 5% is considered for dynamic analysis, shown in Table 4a.

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Details of "Analysis Settings"	
⊞ Step Controls	
⊞ Options	
⊞ Advanced	
⊞ Output Controls	
⊟ Damping Controls	
Eqv. Damping Ratio From...	No
Damping Define By	Damping Ratio
Damping Ratio	5.e-002
⊞ Analysis Data Management	

Table 4a. Damping properties for harmonic analysis

Harmonic analysis of the model provides the response of a system for any sinusoidal loading. Such analyses for a range of frequencies are called frequency response analysis. Plots of different quantities (displacement, velocity, etc.) vs such frequency ranges are called frequency response functions (FRF).

Harmonic analysis is done between 0 to 6.6 Hz frequencies with 100 intervals, shown in Table 4b.

Details of "Analysis Settings"	
⊞ Step Controls	
⊟ Options	
Frequency Spacing	Linear
Range Minimum	0. Hz
Range Maximum	6.6 Hz
Solution Intervals	100
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
On Demand Expansion	No
Store Results At All Frequencies	Yes

Table 4b. Harmonic analysis settings

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5. Results

5.1. Modal Analysis

Table 5 and Figure-7a-7l shows first 50 natural frequencies and first 6 mode shapes. To verify the safe condition of the system at these frequencies, harmonic response analyses are to be performed. As soil is taken in the consideration of the analyses, mass participation considers heavy static mass of the soil along with vibrating modes of the structure. Hence, checking of accumulated mode participation ratio cannot be the effective measure.

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Cumulative Effective Mass Fraction

Mode	Frequency [Hz]	X Direction	Y Direction	Z Direction	Rotation X	Rotation Y	Rotation Z
1	7.2935	9.3669e-003	7.1719e-008	1.2446e-006	7.4244e-008	9.6324e-003	1.2146e-005
2	7.639	9.3669e-003	9.9018e-003	1.4628e-006	1.0178e-002	9.6325e-003	1.7544e-004
3	11.574	9.3674e-003	1.0066e-002	1.6135e-006	1.0347e-002	9.633e-003	1.1474e-003
4	24.061	0.16314	9.0155e-002	1.5002e-003	9.0626e-002	0.16419	1.4411e-003
5	24.103	0.6201	0.12178	4.0461e-003	0.12233	0.62392	1.5067e-003
6	24.457	0.62066	0.52821	3.2391e-002	0.53119	0.62446	1.5203e-003
7	24.646	0.62425	0.71856	7.4935e-002	0.72272	0.62801	1.8018e-003
8	26.213	0.70537	0.71863	7.5475e-002	0.72279	0.70967	3.615e-003
9	27.123	0.70548	0.71863	7.5485e-002	0.72279	0.70978	0.77882
10	29.17	0.70548	0.71948	7.5532e-002	0.72364	0.70978	0.77941
11	30.662	0.70549	0.76306	9.6584e-002	0.76739	0.70979	0.78222
12	30.814	0.7055	0.76469	0.23744	0.76904	0.7098	0.78227
13	31.267	0.70559	0.76596	0.42586	0.77031	0.70989	0.78231
14	32.08	0.70559	0.80025	0.4261	0.80543	0.70989	0.78361
15	32.627	0.74163	0.80026	0.42611	0.80544	0.74714	0.7837
16	34.248	0.74167	0.82432	0.43017	0.82827	0.74717	0.78384
17	34.402	0.75364	0.82448	0.75031	0.82842	0.75845	0.78391
18	34.733	0.80086	0.82453	0.83253	0.82847	0.80308	0.78416
19	35.252	0.80108	0.82482	0.83284	0.82872	0.80329	0.78686
20	35.544	0.80908	0.82499	0.8368	0.82887	0.81032	0.78717
21	35.756	0.80911	0.83763	0.838	0.84023	0.81035	0.79915
22	35.941	0.80942	0.84883	0.83805	0.85029	0.81063	0.81151
23	36.1	0.80944	0.84958	0.84065	0.85097	0.81065	0.81156
24	36.214	0.80949	0.84964	0.84313	0.85103	0.81069	0.81565
25	37.416	0.8096	0.84967	0.84313	0.85105	0.81083	0.92076
26	37.916	0.80993	0.84972	0.84331	0.85112	0.81119	0.92096
27	38.298	0.80996	0.84973	0.84338	0.85112	0.81124	0.92275
28	38.573	0.80997	0.84985	0.84404	0.85123	0.81124	0.96592
29	39.202	0.81015	0.85017	0.84648	0.85154	0.81141	0.96598
30	39.659	0.81057	0.85018	0.89914	0.85155	0.81184	0.96714
31	39.766	0.81164	0.85023	0.89963	0.8516	0.813	0.98957
32	40.01	0.81265	0.85024	0.90107	0.8516	0.81428	0.99395
33	40.586	0.81344	0.85054	0.94016	0.8519	0.81509	0.99409
34	40.956	0.81345	0.85082	0.9403	0.85218	0.8151	0.99483
35	41.016	0.81345	0.85664	0.94243	0.85808	0.8151	0.99506
36	41.355	0.81412	0.85742	0.96427	0.85874	0.81574	0.99506
37	41.437	0.82807	0.85745	0.96456	0.85876	0.83032	0.99506
38	41.625	0.82808	0.86911	0.9687	0.86959	0.83034	0.99508
39	42.221	0.83519	0.87011	0.97336	0.87061	0.83691	0.99508
40	42.445	0.83551	0.88033	0.97396	0.8816	0.83722	0.99509
41	42.981	0.84226	0.88033	0.97407	0.8816	0.84361	0.99512
42	43.268	0.84227	0.88165	0.97509	0.88301	0.84362	0.99514
43	43.852	0.84426	0.88166	0.97542	0.88301	0.84571	0.99517
44	44.43	0.84461	0.88511	0.97553	0.88642	0.84606	0.99599
45	44.549	0.84675	0.90079	0.9763	0.90178	0.84814	0.99607
46	44.746	0.84704	0.91479	0.9763	0.91561	0.84842	0.9965
47	45.083	0.95074	0.91524	0.97739	0.91605	0.95098	0.99651
48	45.215	0.9606	0.98546	0.97812	0.9856	0.96077	0.99928
49	45.529	0.96208	0.98546	0.98456	0.98561	0.96221	0.99947
50	45.746	1.	1.	1.	1.	1.	1.

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Table 5a. First 50 Natural Frequencies of Primary Crusher with Soil layers: Unloaded

Cumulative Effective Mass Fraction

Mode	Frequency [Hz]	X Direction	Y Direction	Z Direction	Rotation X	Rotation Y	Rotation Z
1	6.7195	9.8935e-003	4.5388e-008	9.2832e-007	4.6314e-008	1.0182e-002	1.2764e-005
2	7.1756	9.8935e-003	1.0734e-002	1.2398e-006	1.1042e-002	1.0183e-002	2.3331e-004
3	11.474	9.8938e-003	1.0797e-002	1.4284e-006	1.1108e-002	1.0183e-002	1.1593e-003
4	21.499	9.9686e-003	1.3963e-002	4.0506e-003	1.425e-002	1.0255e-002	1.3362e-003
5	22.034	1.0372e-002	1.5277e-002	4.0935e-002	1.5567e-002	1.0639e-002	1.3989e-003
6	23.339	0.21882	1.5282e-002	4.2639e-002	1.5572e-002	0.22031	1.4169e-003
7	24.462	0.21888	0.72589	4.2678e-002	0.73002	0.22036	1.4365e-003
8	24.747	0.70994	0.72595	4.3292e-002	0.73008	0.71414	1.5782e-003
9	27.12	0.70995	0.72595	4.3297e-002	0.73008	0.71415	0.7788
10	29.169	0.70995	0.72682	4.3335e-002	0.73095	0.71415	0.77939
11	30.633	0.70997	0.75687	0.16148	0.76112	0.71416	0.78146
12	30.715	0.70998	0.77242	0.30382	0.77674	0.71417	0.78231
13	31.199	0.71002	0.77365	0.44695	0.77799	0.71421	0.78235
14	32.08	0.71002	0.80821	0.44714	0.81338	0.71421	0.78366
15	32.612	0.74704	0.80823	0.44715	0.81339	0.75248	0.78375
16	34.222	0.74708	0.82858	0.51854	0.83268	0.75252	0.7839
17	34.286	0.74986	0.83332	0.79657	0.83718	0.75514	0.7839
18	34.674	0.80753	0.83338	0.81301	0.83723	0.80956	0.78423
19	35.25	0.80771	0.8337	0.81314	0.83751	0.80973	0.78705
20	35.524	0.81482	0.83397	0.81389	0.83775	0.81595	0.78758
21	35.721	0.81482	0.84679	0.81499	0.84925	0.81595	0.7977
22	35.903	0.81507	0.85676	0.815	0.85821	0.81617	0.80995
23	36.099	0.81509	0.8573	0.81775	0.85869	0.81619	0.81
24	36.212	0.81512	0.85735	0.82017	0.85873	0.81622	0.81345
25	37.377	0.81537	0.85738	0.82018	0.85876	0.8165	0.91958
26	37.888	0.81583	0.8575	0.8206	0.85889	0.817	0.92045
27	38.296	0.81588	0.8575	0.82093	0.85889	0.81706	0.92195
28	38.554	0.81589	0.85765	0.82351	0.85903	0.81706	0.96279
29	39.135	0.81621	0.85807	0.83289	0.85943	0.81736	0.96415
30	39.552	0.81631	0.85808	0.91359	0.85944	0.81747	0.96553
31	39.705	0.81768	0.85823	0.91415	0.85958	0.81893	0.98975
32	39.985	0.81887	0.85823	0.91586	0.85958	0.8204	0.99364
33	40.414	0.81923	0.85849	0.95348	0.85983	0.82081	0.99411
34	40.841	0.81926	0.86833	0.95505	0.86952	0.82084	0.99416
35	40.961	0.81926	0.86861	0.95509	0.8698	0.82084	0.99512
36	41.271	0.81927	0.86875	0.96735	0.86989	0.82085	0.99512
37	41.434	0.83397	0.86879	0.96735	0.86992	0.83615	0.99512
38	41.509	0.83441	0.87727	0.97074	0.87766	0.83662	0.99513
39	42.074	0.84381	0.87761	0.97256	0.87801	0.84529	0.99513
40	42.437	0.84399	0.88922	0.9728	0.89042	0.84546	0.99514
41	42.889	0.84886	0.88923	0.97332	0.89042	0.85009	0.99519
42	43.231	0.84888	0.89028	0.9745	0.89156	0.85011	0.99521
43	43.819	0.85004	0.8903	0.97497	0.89158	0.85137	0.99524
44	44.42	0.85137	0.89112	0.97564	0.89239	0.85267	0.99593
45	44.496	0.85393	0.90886	0.97712	0.90978	0.85516	0.99613
46	44.746	0.85425	0.92321	0.97713	0.92394	0.85547	0.99656
47	44.962	0.92342	0.92332	0.98088	0.92406	0.92376	0.99658
48	45.169	0.93514	0.97793	0.98142	0.97815	0.9354	0.99889
49	45.445	0.94449	0.97795	0.98907	0.97818	0.94473	0.99909
50	45.688	1.	1.	1.	1.	1.	1.

Table 5b. First 50 Natural Frequencies of Primary Crusher with Soil layers: Loaded

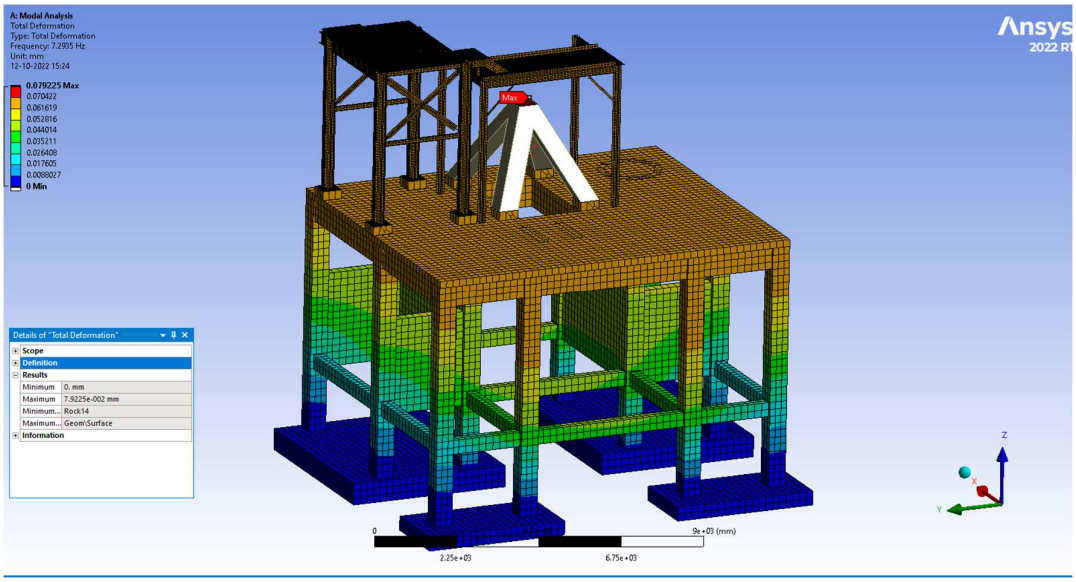


Figure 7a. First Mode Shapes of Primary Crusher: Unloaded.

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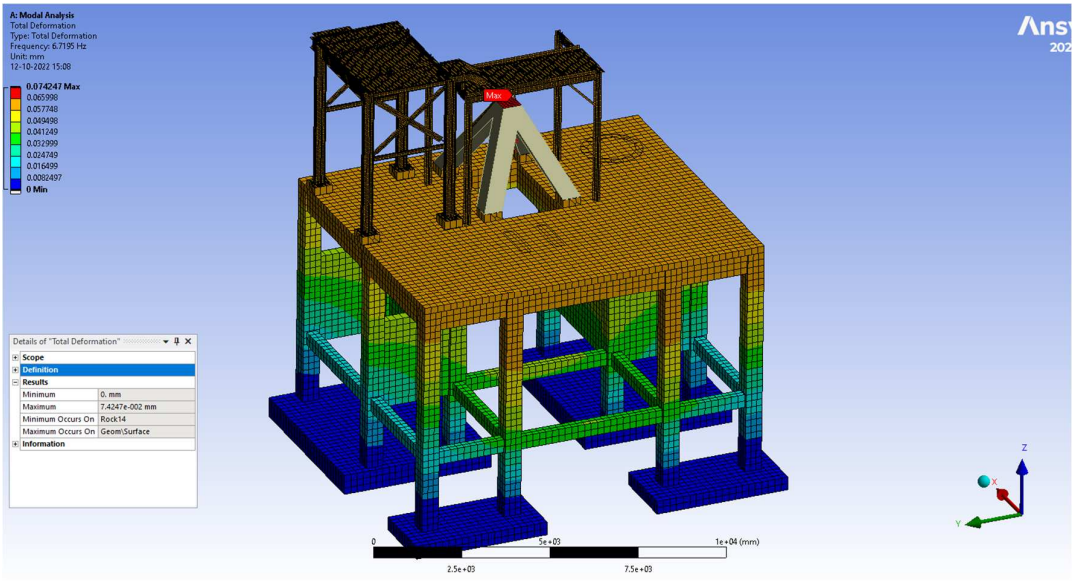


Figure 7b. First Mode Shapes of Primary Crusher: Loaded

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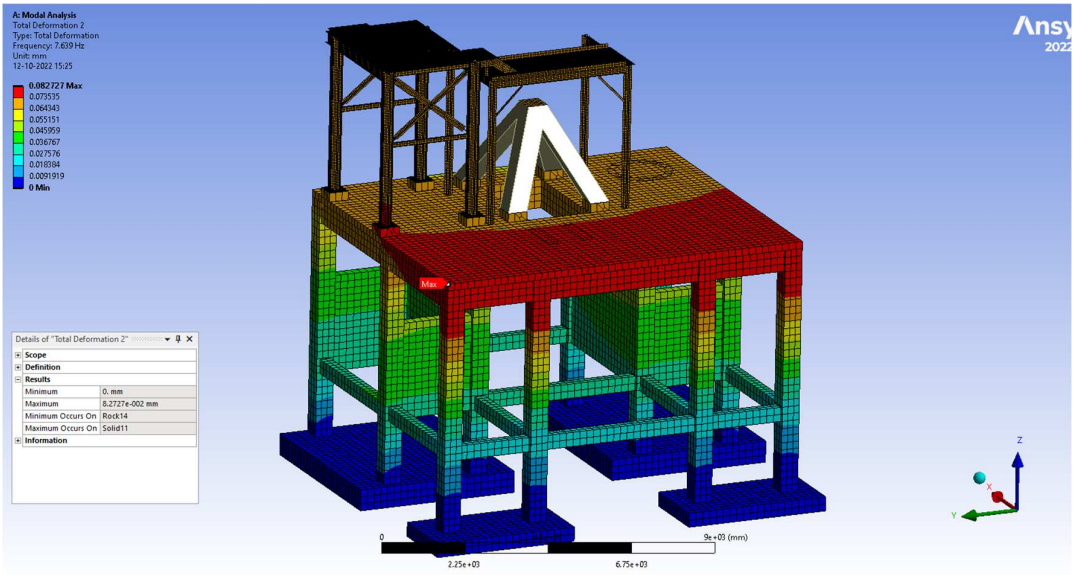
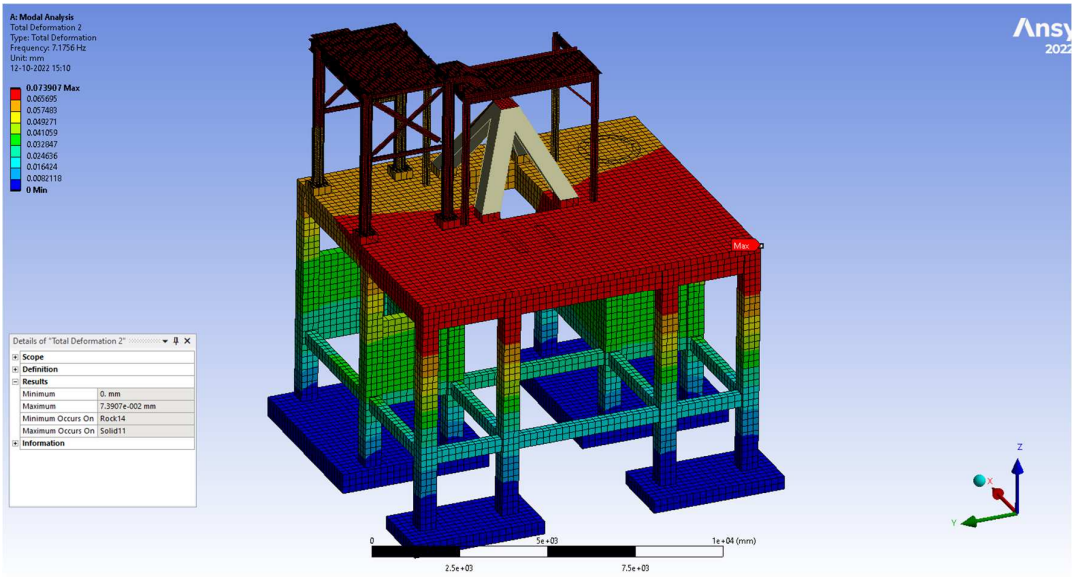


Figure 7c. Second Mode Shapes of Primary Crusher: Unloaded.



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Figure 7d. Second Mode Shapes of Primary Crusher: Loaded

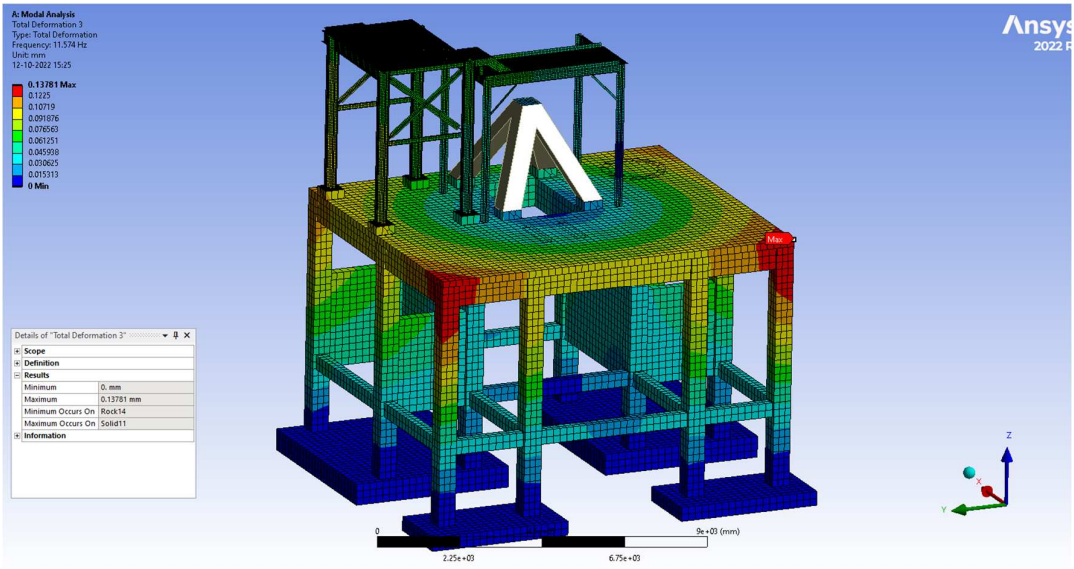


Figure 7e. Third Mode Shapes of Primary Crusher: Unloaded.

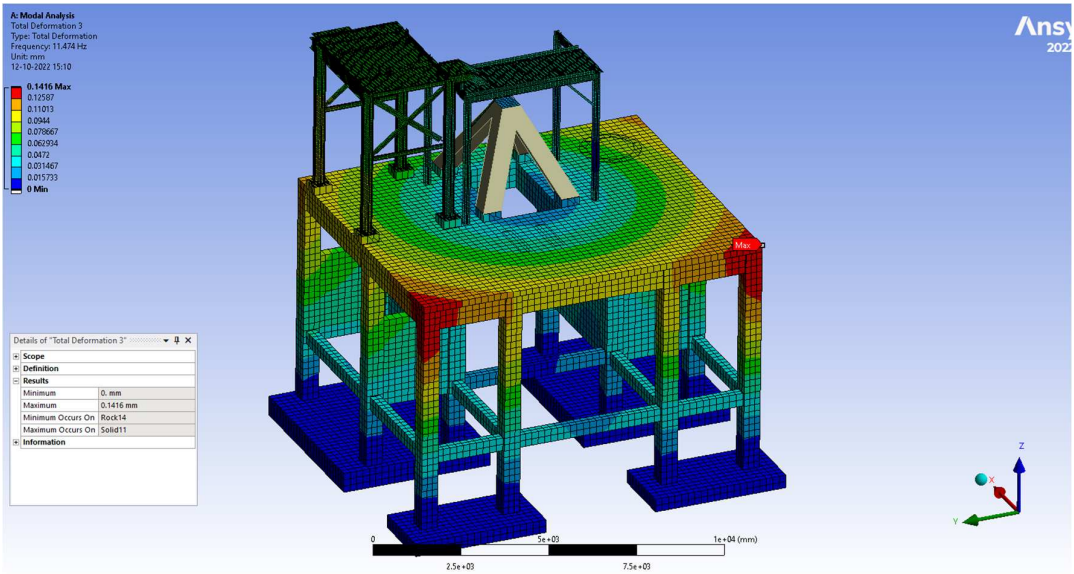


Figure 7f. Third Mode Shapes of Primary Crusher: Loaded

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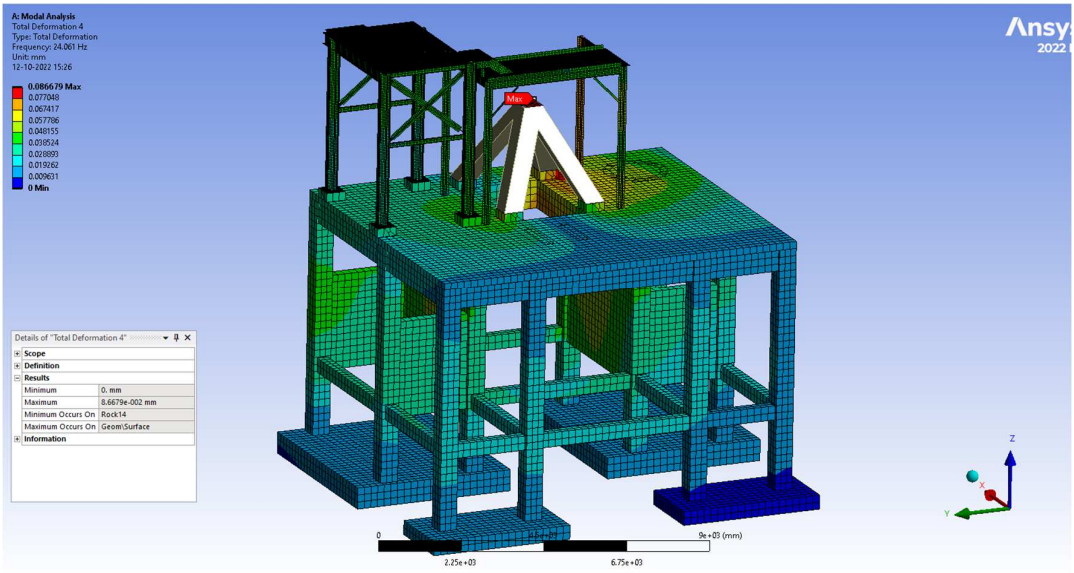


Figure 7g. Fourth Mode Shapes of Primary Crusher: Unloaded

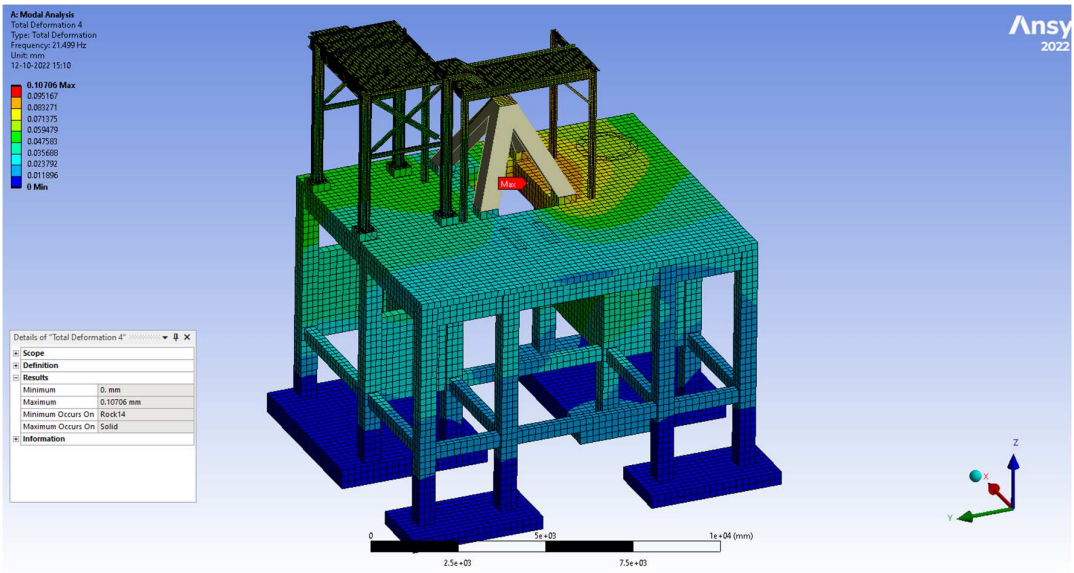


Figure 7h. Fourth Mode Shapes of Primary Crusher: Loaded

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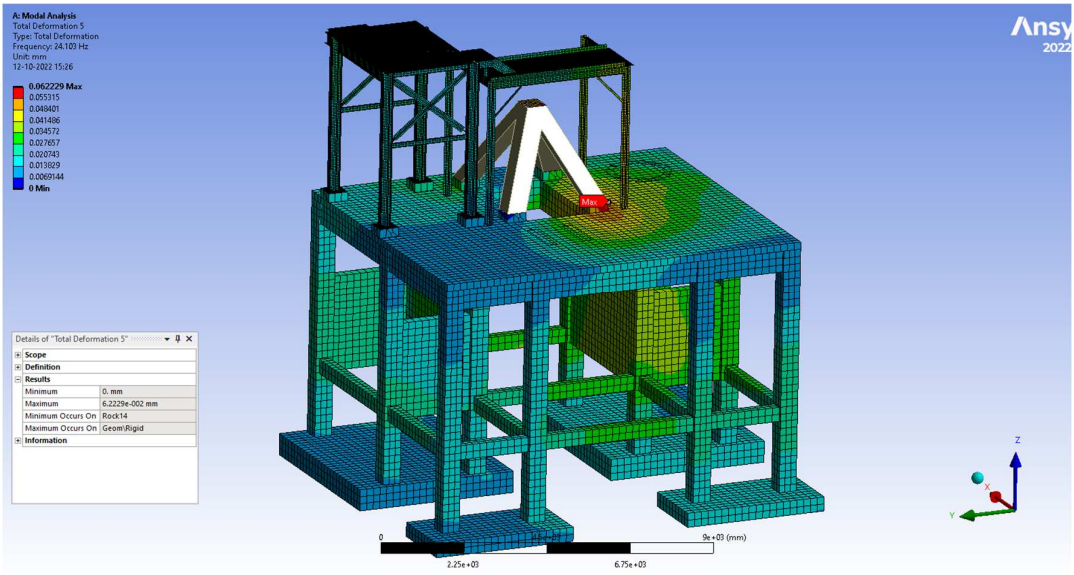


Figure 7i. Fifth Mode Shapes of Primary Crusher: Unloaded

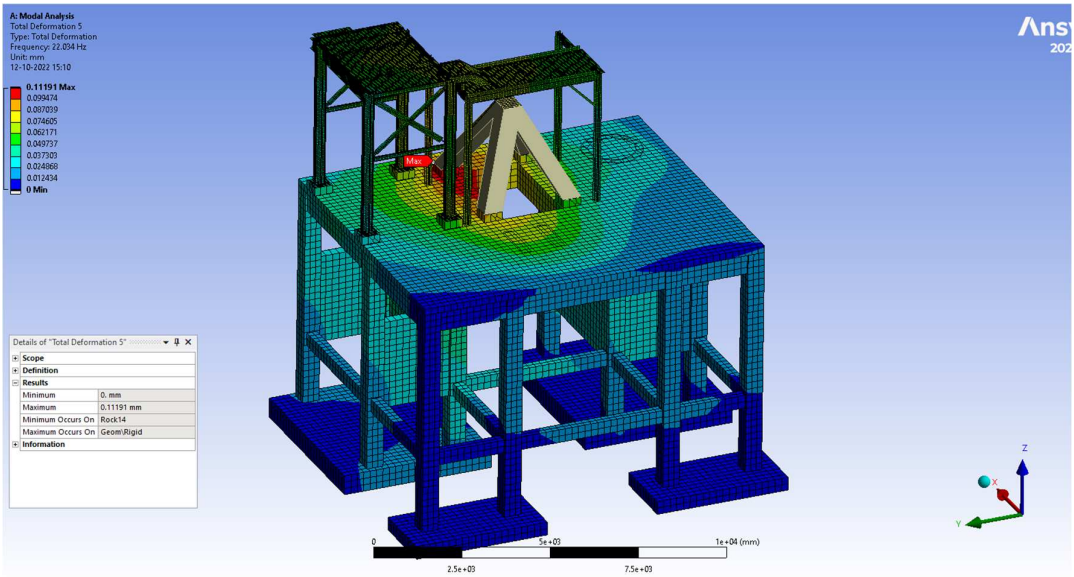


Figure 7j. Fifth Mode Shapes of Primary Crusher: Loaded

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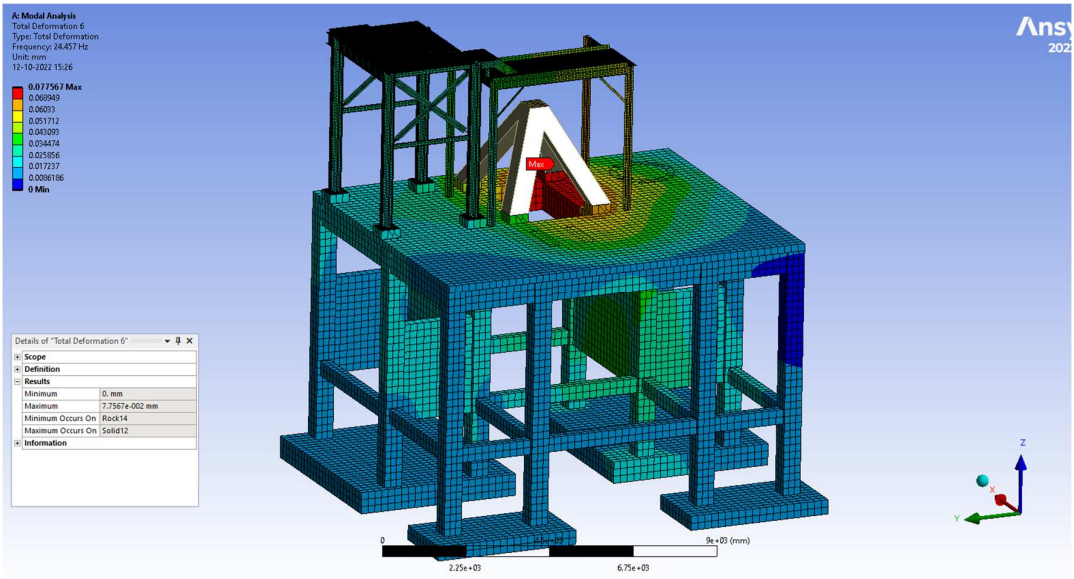


Figure 7k. Sixth Mode Shapes of Primary Crusher: Unloaded

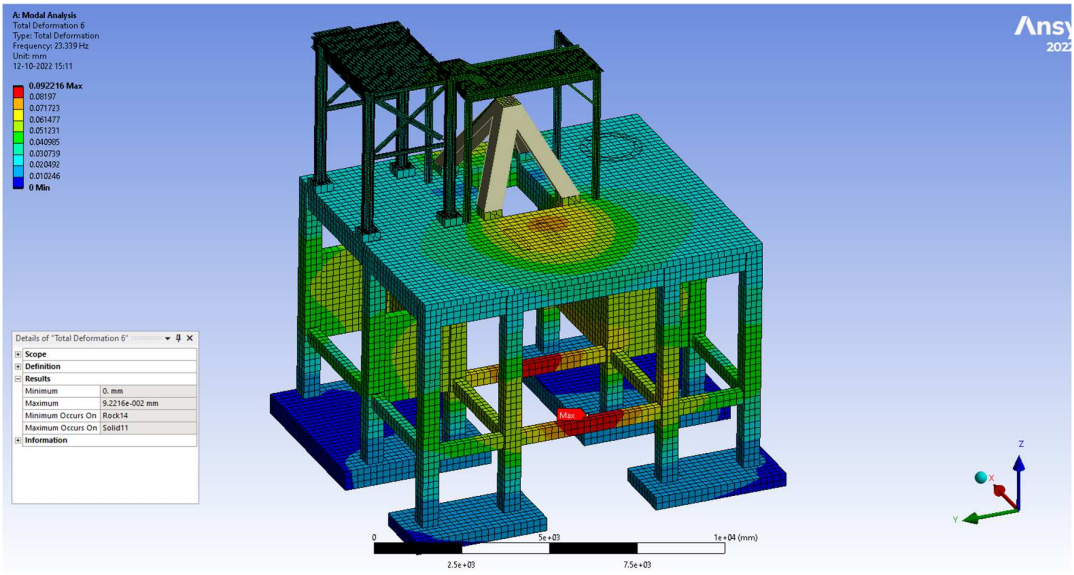


Figure 7l. Sixth Mode Shapes of Primary Crusher: Loaded

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5.2. Harmonic Analysis for Excitation Load Condition 1

5.2.1. Relationships between different terms

Relationship between peak velocity vs peak displacement vs peak accelerations is:

Peak velocity = 2 * PI * frequency * peak displacement ……. Equation (1)

Peak acceleration = 2 * PI * frequency * peak velocity ……. Equation (2)

Where for 3.3 Hz frequency, 2 * PI * 3.3 = 20.73451 rad/sec

Relationship between peak velocities vs R. M. S. velocities is:

R. M. S. velocity = peak velocity / Sqrt (2) ……. Equation (3)

5.2.2. Peak Total Velocity of Concrete Structure, Unloaded Condition: (3.3 Hz)

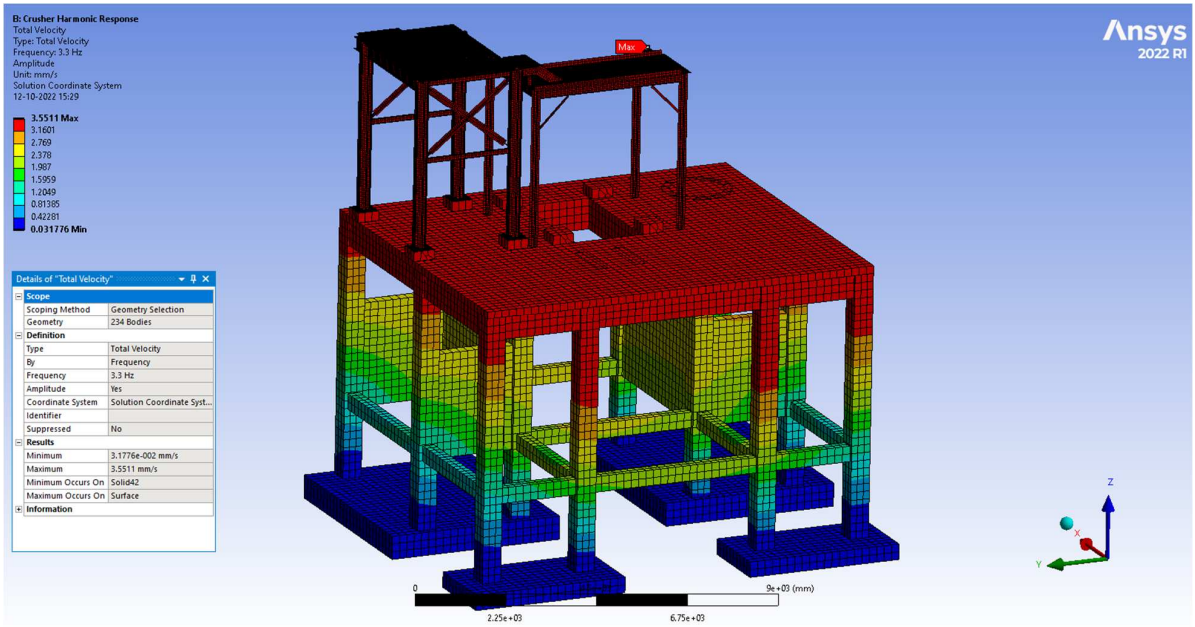


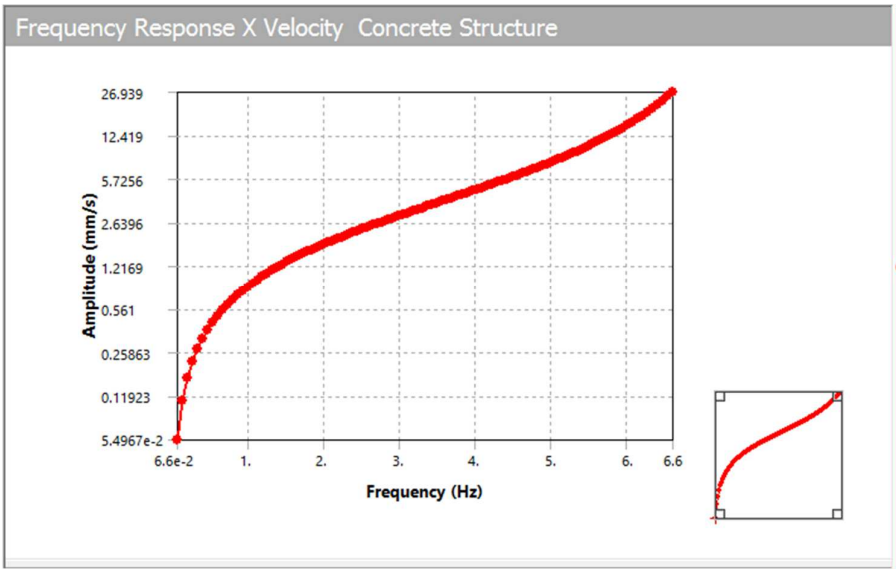
Figure 8a : Maximum peak total velocities among different Loadings

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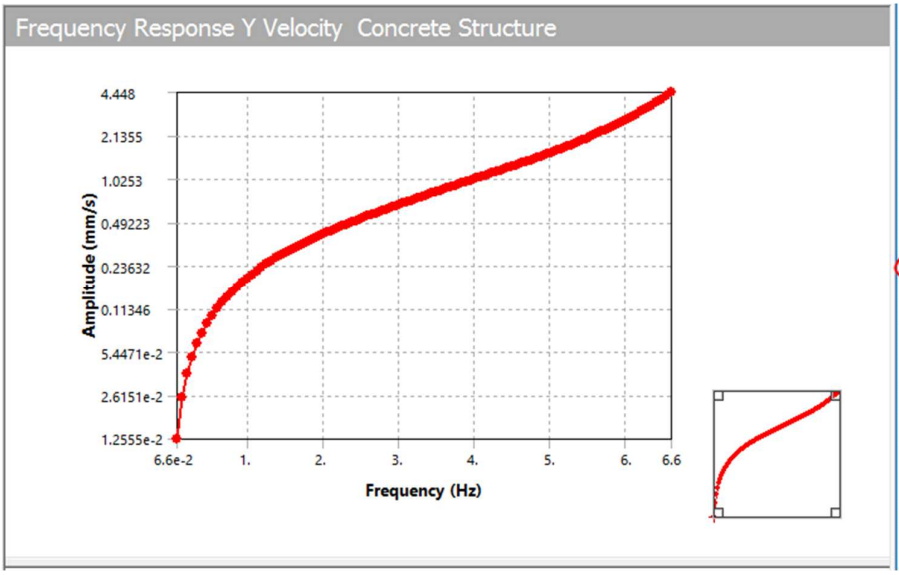
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5.2.3. FRF for Peak Velocity of Concrete Structure, Unloaded Condition: (0-6.6 Hz)

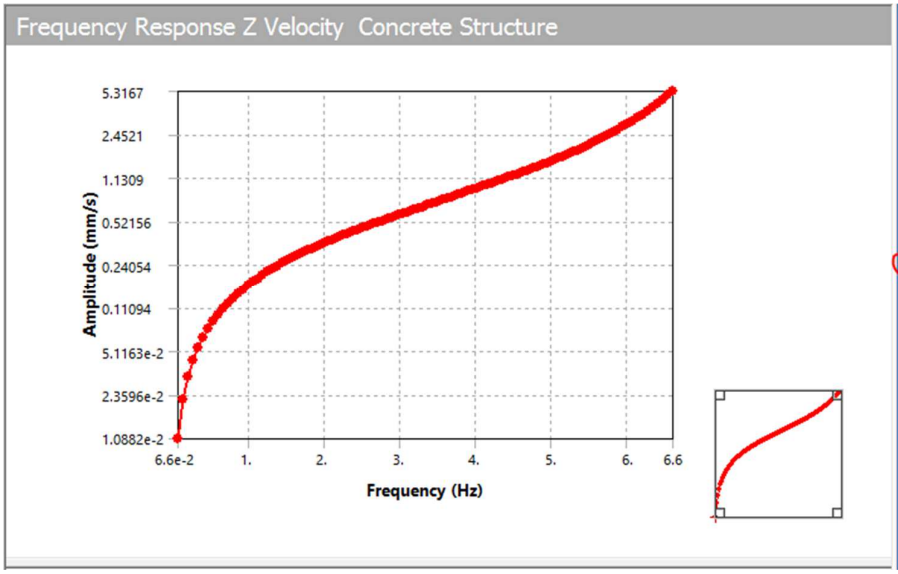


	Frequency [Hz]	Amplitude [mm/s]
40	2.64	2.5243
41	2.706	2.607
42	2.772	2.6915
43	2.838	2.7778
44	2.904	2.8661
45	2.97	2.9564
46	3.036	3.0489
47	3.102	3.1436
48	3.168	3.2408
49	3.234	3.3404
50	3.3	3.4428
51	3.366	3.5479
52	3.432	3.6559
53	3.498	3.7671
54	3.564	3.8815
55	3.63	3.9995
56	3.696	4.121
57	3.762	4.2465
58	3.828	4.3761
59	3.894	4.51
60	3.96	4.6485
61	4.026	4.7919
62	4.092	4.9406
63	4.158	5.0947



	Frequency [Hz]	Amplitude [mm/s]
40	2.64	0.56531
41	2.706	0.58317
42	2.772	0.60136
43	2.838	0.61989
44	2.904	0.63879
45	2.97	0.65806
46	3.036	0.67774
47	3.102	0.69782
48	3.168	0.71835
49	3.234	0.73933
50	3.3	0.7608
51	3.366	0.78277
52	3.432	0.80527
53	3.498	0.82833
54	3.564	0.85198
55	3.63	0.87624
56	3.696	0.90116
57	3.762	0.92676
58	3.828	0.95308
59	3.894	0.98016
60	3.96	1.008
61	4.026	1.0368
62	4.092	1.0664
63	4.158	1.097

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	Frequency [Hz]	Amplitude [mm/s]
40	2.64	0.50003
41	2.706	0.51642
42	2.772	0.53316
43	2.838	0.55026
44	2.904	0.56776
45	2.97	0.58566
46	3.036	0.60399
47	3.102	0.62277
48	3.168	0.64203
49	3.234	0.66178
50	3.3	0.68207
51	3.366	0.7029
52	3.432	0.72432
53	3.498	0.74635
54	3.564	0.76904
55	3.63	0.79241
56	3.696	0.81651
57	3.762	0.84138
58	3.828	0.86706
59	3.894	0.89361
60	3.96	0.92107
61	4.026	0.9495
62	4.092	0.97896
63	4.158	1.0095

Figure 9a- 9c The FRF of peak directional velocities for different frequencies

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5.2.4. Peak Total Velocity of Concrete Structure, Loaded Condition: (3.3 Hz)

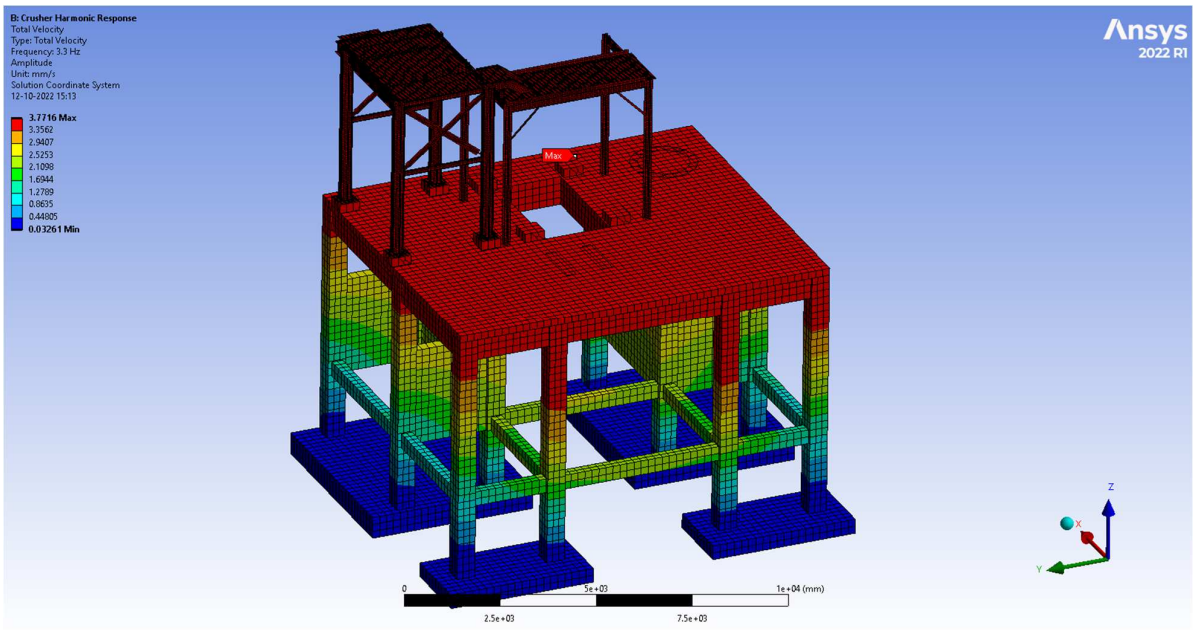
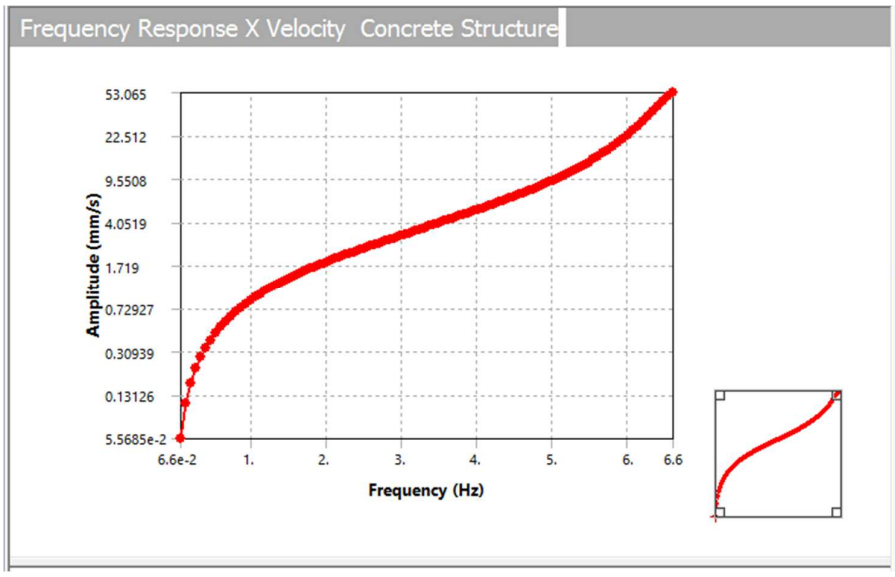


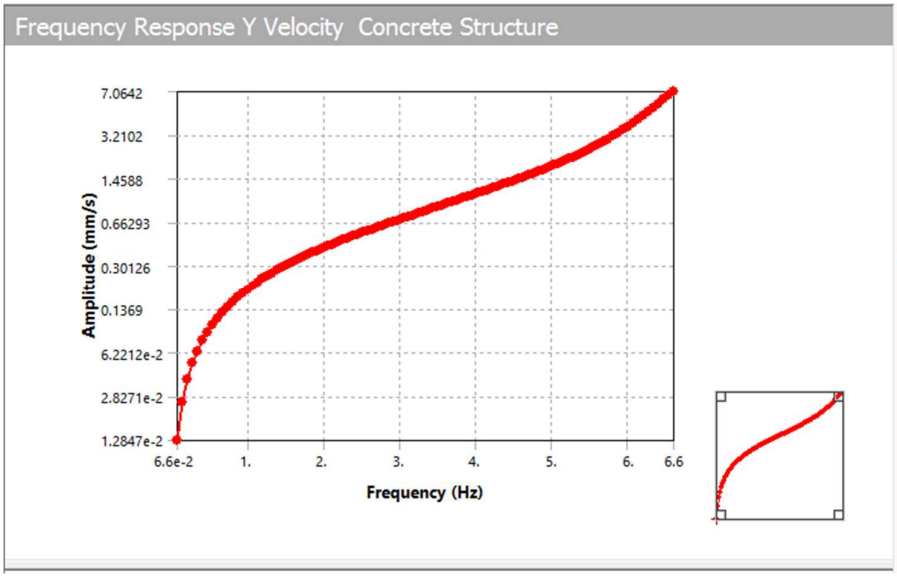
Figure 10a The peak total velocities among different Loadings

5.2.5. FRF for Peak Directional Velocity of Concrete Structure, Loaded Condition: (0.0-6.6 Hz)

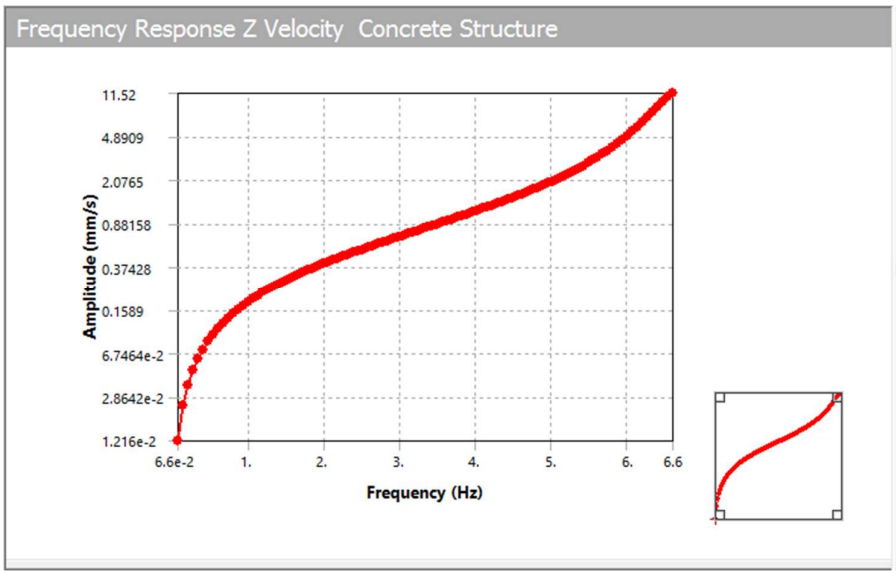


	Frequency [Hz]	Amplitude [mm/s]
40	2.64	2.6287
41	2.706	2.7193
42	2.772	2.8121
43	2.838	2.9074
44	2.904	3.0053
45	2.97	3.106
46	3.036	3.2095
47	3.102	3.3161
48	3.168	3.426
49	3.234	3.5394
50	3.3	3.6564
51	3.366	3.7775
52	3.432	3.9023
53	3.498	4.0317
54	3.564	4.1658
55	3.63	4.3049
56	3.696	4.4493
57	3.762	4.5993
58	3.828	4.7555
59	3.894	4.9181
60	3.96	5.0877
61	4.026	5.2648
62	4.092	5.4499
63	4.158	5.6437

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	Frequency [Hz]	Amplitude [mm/s]
40	2.64	0.59041
41	2.706	0.60979
42	2.772	0.62959
43	2.838	0.64983
44	2.904	0.67054
45	2.97	0.69174
46	3.036	0.71345
47	3.102	0.73571
48	3.168	0.75854
49	3.234	0.78198
50	3.3	0.80605
51	3.366	0.8308
52	3.432	0.85626
53	3.498	0.88247
54	3.564	0.90947
55	3.63	0.93732
56	3.696	0.96607
57	3.762	0.99575
58	3.828	1.0264
59	3.894	1.0582
60	3.96	1.0911
61	4.026	1.1252
62	4.092	1.1606
63	4.158	1.1974



	Frequency [Hz]	Amplitude [mm/s]
40	2.64	0.5741
41	2.706	0.59387
42	2.772	0.61415
43	2.838	0.63497
44	2.904	0.65635
45	2.97	0.67833
46	3.036	0.70095
47	3.102	0.72423
48	3.168	0.74823
49	3.234	0.77298
50	3.3	0.79854
51	3.366	0.82494
52	3.432	0.85224
53	3.498	0.8805
54	3.564	0.90978
55	3.63	0.94014
56	3.696	0.97167
57	3.762	1.0044
58	3.828	1.0385
59	3.894	1.074
60	3.96	1.1111
61	4.026	1.1497
62	4.092	1.1901
63	4.158	1.2324

Figure 11a- 11c shows the FRF of peak directional velocities for different frequencies

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5.2.6. Summary of Results

Table 6a shows the maximum Peak displacement, Velocity, and acceleration in the concrete structure for unloaded & loaded conditions. Table 6b shows the R.M.S. velocity of the structure.

		Total Peak	X Peak	Y Peak	Z Peak
Maximum Velocities (mm/sec)	Unloaded	3.551	3.44	0.760	0.682
	Loaded	3.772	3.66	0.806	0.799
Maximum Displacement (micron)	Unloaded	45.22	43.82	9.68	8.68
	Loaded	48.02	46.60	10.26	10.17
Maximum Acceleration (m/sec^2)	Unloaded	0.279	0.270	0.060	0.054
	Loaded	0.296	0.287	0.063	0.063

Table 6a. Peak Displacement, Velocity, & Acceleration

		Total RMS	X RMS	Y RMS	Z RMS
Maximum Velocities (mm/sec)	Unloaded	2.511	2.434	0.537	0.482
	Loaded	2.667	2.588	0.570	0.565

Table 6b. RMS Velocity of the structure

6. Conclusions

The natural frequency of the system has been high-tuned [6.72Hz] to 2 times of working frequency [3.3 Hz]. The maximum peak vibration level for the x, y and z directions over the excitation forcing frequency 3.3Hz is computed in the finite element model. The absolute maximum R.M.S. velocity level computed for frequency is 2.511 & 2.667 mm/s for unloaded & loaded condition respectively which is within project specified limit of 3.0 mm/s .Peak displacements computed 45.22 & 48.02 microns for unloaded & loaded conditions respectively.

7. References

[1] ANSYS Mechanical Software

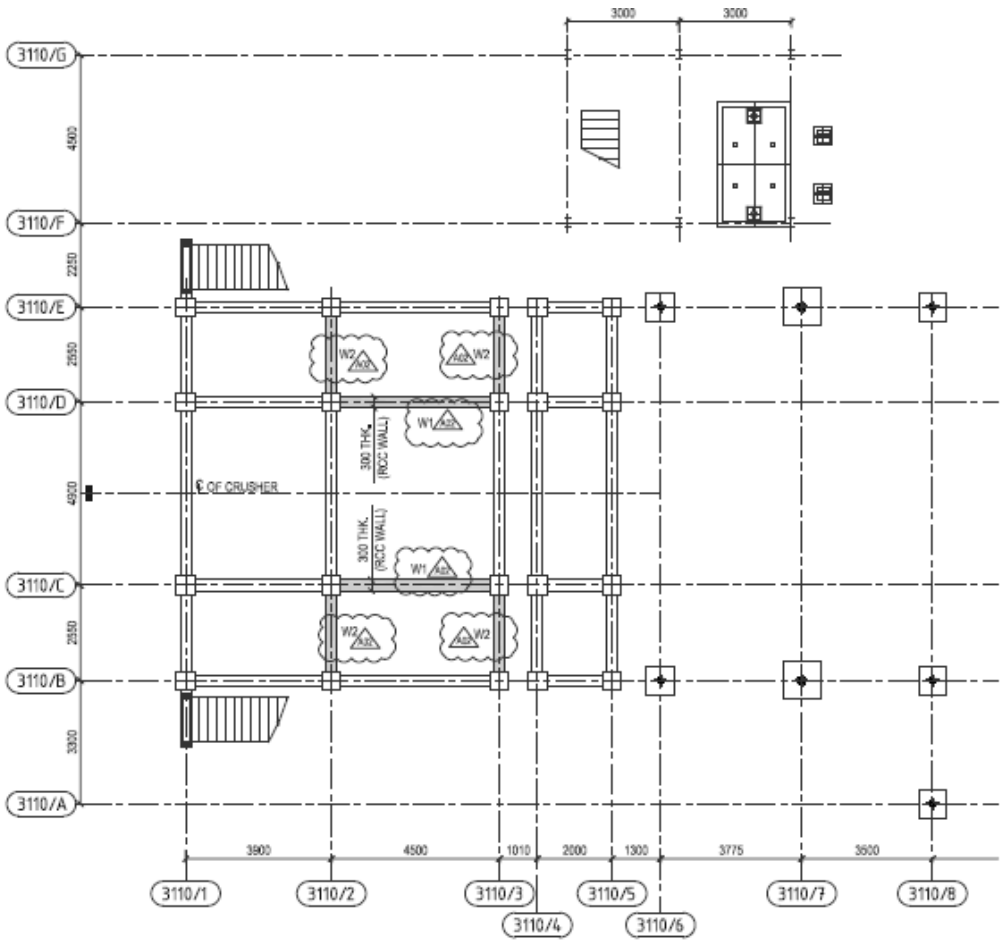
[2] MANSOURAH - MASSARAH GOLD PROJECT GENERAL ARRANGEMENT JAW CRUSHER 3110-CRH-001 (MD - 216 -3110 - EG- ME -GAD -V500-1007) Rev -C01.

[3] MANSOURAH - MASSARAH GOLD PROJECT GENERAL ARRANGEMENT PRIMARY CRUSHER JIB CRANE PLAN & ELEVATION VIEW (MD - 216 -3110 - EG- ME -GAD -V557-1001) Rev -C02.

[4] MD-216-3110-EG-CV-CAL-1004-C02

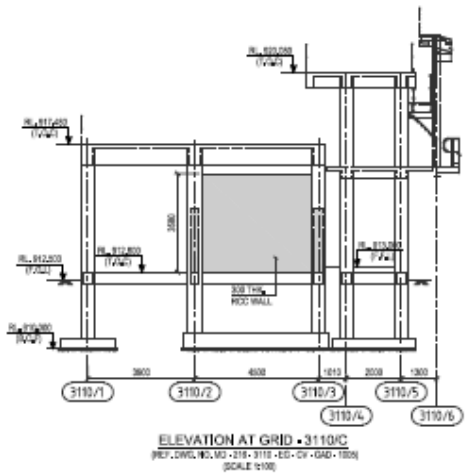
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8. Appendix 1: Modified Primary Crusher supporting Structure

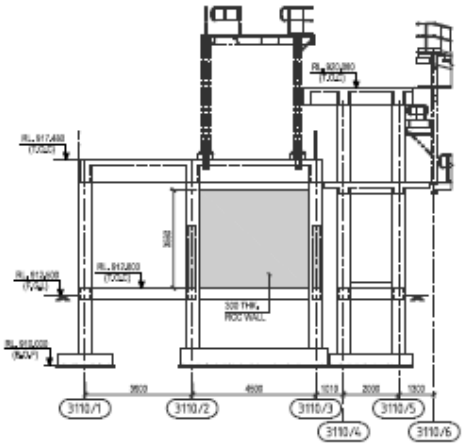


TIE BEAM AND NEW RCC WALL LAYOUT AT RL. 912,800 (T.O.C)
(REF. DWG. NO. MD - 216 - 3110 - EG - CV - GAD - 1004)
(SCALE 1:100)

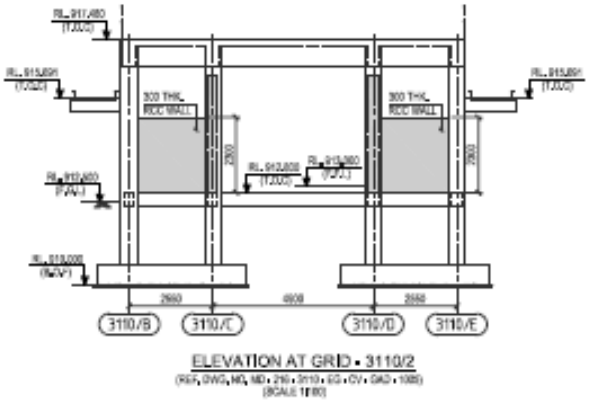
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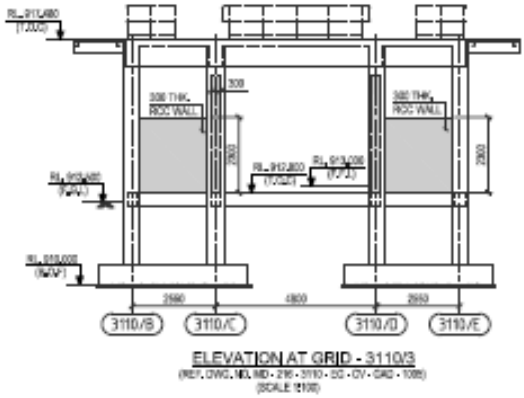
ELEVATION AT GRID - 3110/C



ELEVATION AT GRID - 3110/D



ELEVATION AT GRID - 3110/2



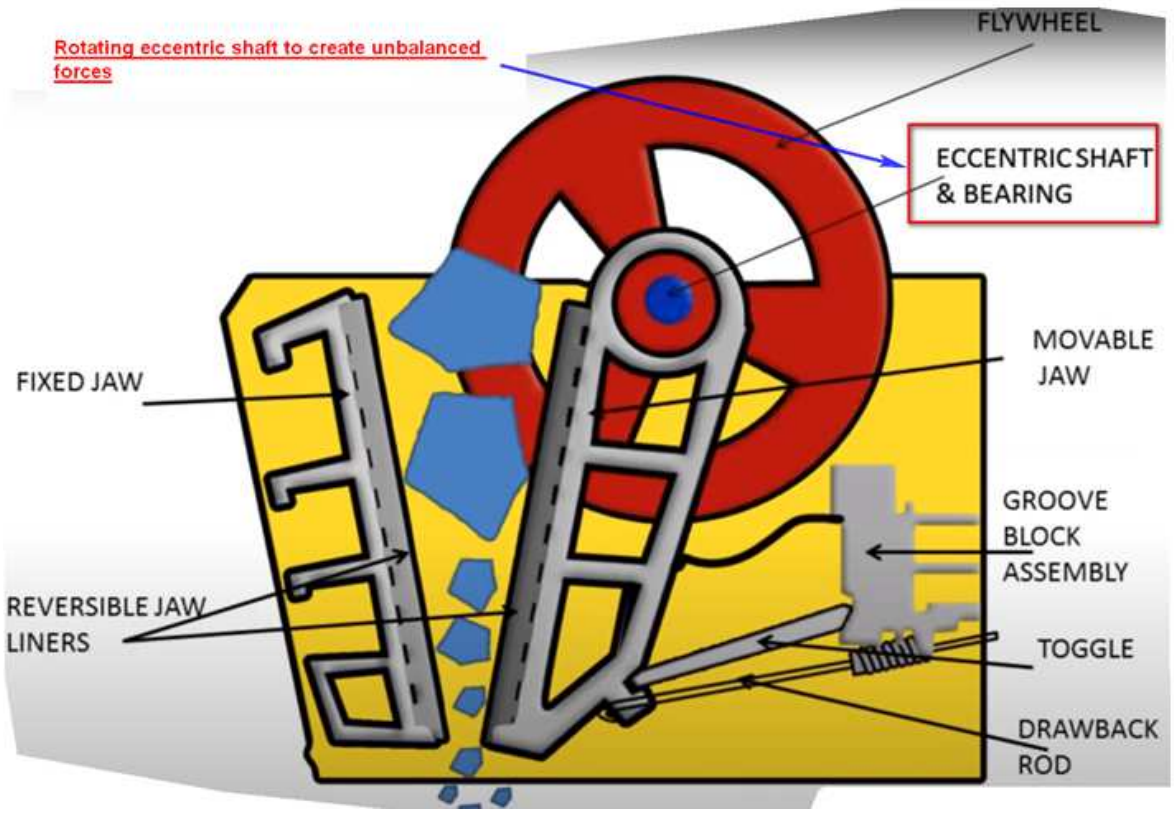
ELEVATION AT GRID - 3110/3

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9. Appendix 2: Typical Figure of the Forcing Mechanism



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